

AEROVANTI SKYPOINTS

Project Documentation

ETL · data model · DAX · report · analysis walkthrough
Fictional company · synthetic seeded data · generated 2026-09-02

00 — Scope & Scaffold (Phase 0)

Project: AeroVanti SkyPoints — Redemption & Retention Analysis **Fictional company:** AeroVanti — a Brazilian domestic airline.
Loyalty program: SkyPoints (miles-based earn/burn, tiered). **Date locked:** 2026-09-02

1. Premise

SkyPoints members accumulate miles but rarely redeem them: roughly 40% of issued points sit unused as **breakage**, and members who do not redeem within 12 months **lapse** at a materially higher rate. The loyalty team piloted **Flash Redemption** — a lower minimum redemption threshold plus more low-cost reward options — as a **randomized A/B test** on a subset of active members. This project builds the read-out to decide on full rollout, plus a lighter dashboard on program health.

2. Locked scope decisions

Dimension	Decision
Deliverable language	English (all docs, code, reports). Working chat in Portuguese.
Priority	Phase 10 (data analysis + A/B test) is the primary deliverable. Power BI track is secondary and scoped down.
PBI environment	<code>.pbip</code> / PBIR, built from code (<code>gen_semantic_model.py</code>), theme via <code>pbir</code> CLI.
Curated storage	CSV and Parquet in <code>data/curated/</code> .

Dimension	Decision
Market / currency	Brazil only, BRL only. No multi-currency, no expansion modelling.
Data window	24 months: 2024-09-01 → 2026-08-31. CY = last 12 months (2025-09 → 2026-08), PY = prior 12.
Experiment window	12 weeks in program months 19–21: 2026-03-02 → 2026-05-24. Post-period Jun–Aug 2026 reserved for lapse/retention outcomes.
ETL / data-quality depth	Reduced. ~5–6 simulated source systems, ~6 core data-quality issue classes (see §5).
Program scale	~120,000 enrolled members , ~65,000 active (earn or flight in last 12 months).
Experiment scale	~15,000 members in the pilot, 50/50 split (~7,500 per arm).
Power BI track	Playbook "lite": semantic model as code + <code>lean_theme.json</code> + build script, 4 pages , few formatting passes.

3. Phase-1 report pages (lean, 4 pages)

1. **Executive Summary** — program health + the A/B test headline result.
2. **Member Tiers & Value** — points earned / redeemed, tier movement, breakage.
3. **A/B Test Results** — Flash Redemption experiment, dedicated page.
4. **Redemption & Engagement Funnel** — catalog behaviour and lapse risk.

4. A/B test design (locked at Phase 0, detailed in Phase 1)

Element	Decision
Unit of randomization	Member.
Split	50/50 , treatment vs control.
Eligibility	Members with an earn event or a flight in the trailing 12 months and not lapsed as of the start of month 19 (2026-03-01).
Assignment	Stratified by tier (Blue / Silver / Gold / Platinum) so arms are balanced on tier.
Treatment	Flash Redemption: minimum redemption threshold lowered (e.g. 10,000 → 3,500 pts) and an expanded low-cost reward catalog (gift cards, seat selection, lounge passes, partner vouchers).
Control	Existing thresholds and catalog.
Primary metric	Redemption rate = share of members with ≥1 redemption during the 12-week window.
Guardrail metrics	Revenue per member (BRL), point-liability cost per member (BRL), churn / lapse rate (post-window), average redemption value.
Analyses required	SRM / randomization check; stated hypothesis + MDE / power rationale; significance test with effect size + CI; novelty-effect check across the window; heterogeneity by tier (top vs entry).

Underlying true effect: deliberately decided during Phase 3 (ETL) and baked into the generated data with plausible noise, novelty decay, and tier heterogeneity. The true value lives only in `et1/config.py`; the Phase 10 analysis must detect and quantify it as if blind.

5. Data-quality issues to inject (reduced set)

1. Multiple date formats + Excel serial numbers across source exports.
2. Money as strings (`"R$ 1.234,56"` , `"1234.56"` , thousands separators).
3. Mixed text encodings / mojibake in member names and city fields.
4. Duplicate rows in earn/redemption event exports.
5. Missing master rows (member dimension rows absent for some events → referential gap).
6. Snapshot-vs-ledger drift: a tier-status snapshot table that disagrees with the tier-movement ledger.

6. Out of scope (explicit)

- Multi-currency, FX, international routes, codeshare partners' own accrual.
- Fraud detection, points-trading abuse.
- Real-time / streaming data; all batch.
- Marketing-channel attribution beyond the experiment.
- Fare pricing / revenue management modelling (revenue is an input, not modelled).
- Mobile-app telemetry / clickstream.

7. Resolved-assumptions log

#	Assumption	Resolution
A1	"Active member" definition	Earn event or flight in trailing 12 months.
A2	Program size	120k enrolled; growth curve modelled over the 24 months (Phase 1).
A3	Tiers	4: Blue (entry), Silver, Gold, Platinum. Qualification by trailing-12m miles or segments (Phase 1).
A4	Point value	~BRL 0.025 per point redeemed value; liability booked at issuance (Phase 1 to finalize).
A5	Breakage baseline	~40% of issued points never redeemed (program-to-date).
A6	Lapse definition	No earn and no redemption for 12 consecutive months.
A7	Experiment enrollment	Stratified random by tier, 15k members, 50/50.
A8	Currency formatting in report	pt-BR display acceptable? To confirm in Phase 5 (English portfolio → likely switch display units).

8. Folder scaffold (created)

```
AeroVanti-SkyPoints-Analytics/  
docs/           business context, architecture, data dict, model, DAX, DQ, walkthrough  
etl/           config + pipeline (raw → staging → curated) + dq_checks  
data/  
  raw/         source-faithful messy exports  
  staging/      typed stg_* tables  
  curated/     Kimball star, CSV + Parquet  
  quality/     dq_report.md and fragments  
powerbi/       semantic model as code, theme, build_report.sh, .Report / .SemanticModel  
assets/        logo + page-background generators  
data_analysis/ sql/ python/ findings/ parity/ deliverables/ outputs/ ← PRIMARY  
portfolio/     gallery images (diagram + screenshot sources)  
README.md
```

AeroVanti SkyPoints — Business Context & KPI Framework

Document status: Phase 1 deliverable · authored at project kickoff, before any analysis. **Data generation date:** 2026-09-02 · **Program-history window:** 2024-09-01 → 2026-08-31 (24 full months). **Embedded experiment:** Flash Redemption A/B test, 2026-03-02 → 2026-05-24 (12 weeks, program months 19–21). **Audience:** the AeroVanti loyalty leadership team and the CFO.

This document frames the problem and defines the metrics. It deliberately does **not** state whether Flash Redemption worked, or by how much — that is the job of the read-out (Phase 10). Everything here was knowable *before* the analysis began. The true underlying treatment effect is decided during ETL (Phase 3), baked into the synthetic data with noise, novelty decay and tier heterogeneity, and kept only in `et1/config.py` ; Phase 10 must detect and quantify it as if blind.

1. The situation

AeroVanti is a mid-size Brazilian domestic airline. Its frequent-flyer program, **SkyPoints**, was relaunched two years ago (Sep 2024) with a modern tier structure and a co-branded credit card; this project's 24-month window is exactly the relaunched program's life to date.

Two things are wrong with the program:

- Members hoard points.** Only about one active member in four has *ever* redeemed. Across the program to date, roughly **40% of issued points are expected to expire unredeemed** ("breakage"). A large, slow-growing **point liability** sits on the balance sheet, and the CFO wants it down.
- Non-redeemers leave.** Members who have never redeemed **lapse** (go 12 months with no earn and no redemption) at a materially higher rate than members who have redeemed at least once. Every lapsed member is lost forward flight revenue.

The loyalty team's hypothesis is that a **redemption threshold that is too high** is the common cause: a member with 4,000 points can't do anything with them, never gets the "I got something" moment, disengages, and eventually lapses — while their points age toward expiry. In program months 19–21 the team ran **Flash Redemption** as a **randomized, member-level A/B test**: a lower minimum redemption threshold plus an expanded catalog of low-cost rewards. This project builds the read-out needed to decide on full rollout, plus a lighter dashboard on overall program health.

2. Company profile

Name	AeroVanti (legal entity <i>AeroVanti Linhas Aéreas S.A.</i>)
Headquarters	Campinas (VCP), São Paulo state, Brazil
Business	Domestic scheduled passenger airline — point-to-point + two-hub (VCP, Recife/REC)
Fleet & network	~28 narrowbody aircraft; ~42 domestic routes; ~4.5M passengers/year
Age	Airline ~12 years; SkyPoints relaunched Sep 2024 (legacy program folded in)
Currency	BRL only. Domestic operation, no international routes, no FX in scope.
Fiscal calendar	Calendar year; loyalty reporting is monthly, experiment reporting is weekly
Program scale at window end	~150k cumulative enrolled members; ~65k active (earn or flight in trailing 12 months); outstanding point liability ~R\$14M

SkyPoints positioning

A **coalition-lite** airline program: points are earned on AeroVanti flights (with a tier multiplier), on the **BancoAV SkyPoints Mastercard** co-branded card (the single largest issuance source), and with a handful of hotel / car-rental / retail partners. Points are redeemed mostly for **award flights**, plus a small ancillary catalog (seat selection, extra bag, lounge passes). The relaunch's selling point was "a clear path to a free flight"; the unsolved problem is that most members never get close.

3. Two point currencies

SkyPoints keeps the industry-standard split between a **spendable** currency and a **status** currency. They must not be conflated in the analysis.

Currency	Purpose	Earns on	Expires	Tier multiplier
SkyPoints	Spendable — redeemed for awards. Drives the liability .	Flights, co-brand card, partners	24 months after the month earned , per lot (FIFO burn)	Yes (see §4)
Tier Points (TP)	Status only — determine tier. Never redeemable, never a liability.	Flights only (base fare + 100 TP/segment)	Reset on the member's tier anniversary (rolling 12-month qualification)	No

Because the program relaunched exactly 24 months before window end and SkyPoints lots live 24 months, **the first material wave of point expiry lands in the last ~3 months of the window** — breakage becomes directly observable, not just forecast, right where the data ends.

4. Tier structure

Four tiers. Qualification is on **trailing-12-month Tier Points or qualifying segments**, re-evaluated monthly; a tier, once earned, is held for 12 months before it can be lost.

Tier	Qualify (trailing 12m): TP or segments	SkyPoints earn multiplier (flights)	~Share of active members	Key benefits
Blue	entry — any enrolled member	1.0×	~68%	Earn points; standard fees
Silver	8,000 TP or 4 segments	1.25×	~20%	Free seat selection, priority check-in
Gold	25,000 TP or 12 segments	1.5×	~9%	Free bag, lounge access, waived change fees
Platinum	60,000 TP or 30 segments	2.0×	~3%	Upgrades, guaranteed availability, bonus redemption value

Tier movement is tracked two ways, and the two are allowed to disagree (a deliberate data-quality issue, §12 of the architecture doc): a monthly **tier-status snapshot** table and a **tier-change event ledger**. The curated model rebuilds status *from* the ledger and reconciles the drift.

5. Points economics — earn, burn, breakage, liability

5.1 Earn (SkyPoints issuance)

Source	Rate	~Share of SkyPoints issued	Notes
AeroVanti flights	1.0 SkyPoint per R\$1 base fare (excl. taxes/fees) × tier multiplier; +50% on flexible fare families	~45%	The only source that also earns Tier Points
BancoAV co-brand card	1 SkyPoint per R\$2.50 card spend; large sign-up bonus on approval	~42%	Largest single issuance source ; no flight attached → pure liability with weak redemption pull
Partners (hotels, car rental, retail, fuel)	Varies by partner	~8%	Small, irregular
Promotions / service recovery / bonuses	Ad hoc	~5%	Double-points campaigns, IRROPS goodwill

5.2 Burn (redemption)

Option	Indicative price	Value per point	Available pre-Flash?
Award flight (dynamic pricing + "SkyCash" points-plus-cash)	from ~8,000 SkyPoints one-way short-haul	~R\$0.024 / pt	Yes
Seat selection	1,500 pts	~R\$0.013 / pt	No — added by Flash

Option	Indicative price	Value per point	Available pre-Flash?
Extra checked bag	2,500 pts	~R\$0.014 / pt	No — added by Flash
Lounge day pass	5,000 pts	~R\$0.016 / pt	No — added by Flash
Partner gift card (R\$25 / R\$50)	3,500 / 6,500 pts	~R\$0.008–0.010 / pt	No — added by Flash
Onboard Wi-Fi / bar voucher	1,800 pts	~R\$0.012 / pt	No — added by Flash

Minimum redemption threshold = the balance below which the redemption UI will not transact at all. **Pre-Flash: 10,000 SkyPoints. Flash (treatment): 3,500 SkyPoints**, plus the low-cost catalog above. A large block of Blue members hold 3,500–9,999 points and are **structurally unable to redeem anything** pre-Flash — this is the segment the intervention is aimed at.

Burn is **FIFO** across point lots (oldest first), so any redemption also directly reduces near-term breakage.

5.3 Breakage & liability (CFO deep-dive thread #1)

Term	Definition
Points outstanding	Σ (issued – redeemed – expired), all lots, at a date.
Point liability (BRL)	Points outstanding $\times$ unit cost, where unit cost $\approx$ R\$0.021 / pt (weighted award value) $\times$ (1 – expected breakage) for the deferred-revenue portion. Reported both gross and breakage-adjusted.
Breakage rate	Of points issued in a cohort (issuance month), the share that expire unredeemed. Estimated cohort-based (preferred) and cross-checked against an aggregate roll-forward.
Breakage-adjusted cost	The P&L cost actually recognised; sensitive to the breakage assumption — a sensitivity table is a required Phase-10 artifact.
Liability roll-forward	Opening + issued – redeemed – expired – revaluation = closing, monthly.

Program-to-date breakage is running near **40%**. The relaunch business case assumed it would fall toward **30%** as members learned the program; it has not. Flash Redemption is also a **liability-management** lever: more redemption draws the liability down and converts "will expire" points into engagement — but the low-cost catalog is bought at **cash cost** from partners, so the net financial effect is not obviously positive and must be quantified.

6. Lapse & retention (CFO deep-dive thread #2)

Term	Definition
Active member	≥ 1 earn event or ≥ 1 flight in the trailing 12 months.
Lapsed member (account lapse)	12 consecutive months with no earn and no redemption . A state, re-evaluated monthly. Note: co-brand-card auto-earn keeps most card-holders' accounts "active" on this definition even when they never fly or redeem, so account lapse understates disengagement.
Engagement lapse	12 consecutive months with no flight and no redemption (card-only earn does not count). The retention metric the program actually cares about — separates "still enrolled" from "still flying with us".
Reactivation	A lapsed member with a new earn or redemption.
Redeemer	A member with ≥ 1 completed redemption, ever (or in a stated window).
Redeemer lapse gap	Lapse rate of never-redeemers – lapse rate of redeemers, controlling for tenure, tier and earn activity. The program's core retention hypothesis: this gap is large and at least partly causal .

The experiment is the clean causal test of that hypothesis: if randomly *enabling* more redemption reduces downstream lapse, the lever is real. Because the experiment sits in months 19–21 and the data ends at month 24, we observe **~14 weeks of post-window behaviour** — enough for early-lapse proxies and a 90-day forward-activity read, not enough for a full 12-month lapse outcome. Phase 10 states this horizon limit explicitly.

7. The Flash Redemption A/B test

7.1 Design

Element	Decision
Unit of randomization	Member (single SkyPoints account).
Eligible population	Active (earn or flight in trailing 12m) and not lapsed as of 2026-03-01 . Frozen list; ~58k members.
Sample & allocation	15,000 members, stratified by tier with mild oversampling of Gold/Platinum so the heterogeneity cut has power: Blue 8,000 · Silver 4,000 · Gold 2,200 · Platinum 800. 50/50 treatment/control within each stratum.
Why oversample	Proportional allocation would leave ~450 Platinum members total — no power for a per-tier effect. Oversampling trades a little external validity for a usable heterogeneity read; the overall effect is reported both pooled and post-stratified back to the population tier mix.
Treatment	Minimum redemption threshold 10,000 → 3,500 SkyPoints ; low-cost reward catalog (\$5.2) switched on; one announcement email + in-app banner.
Control	No change: 10,000 threshold, flights-and-ancillaries-only catalog, no announcement.
Window	12 weeks: 2026-03-02 → 2026-05-24 . Post-window observation to 2026-08-31.
Assignment integrity	Flag set once at enrolment; no re-randomization; no mid-flight changes to thresholds for either arm.

7.2 Hypotheses

- **Business hypothesis.** Lowering the threshold and adding low-cost rewards raises the share of members who redeem, which (a) draws down the point liability and future breakage and (b) reduces downstream lapse, **without** a material loss of flight revenue.
- **Statistical hypothesis (primary).** H_0 : redemption rate is equal in the two arms over the window. H_1 : treatment > control. Tested one-sided at $\alpha = 0.05$; a **two-sided 95% CI** on the difference is always reported.

7.3 Metrics

Role	Metric	Definition	Decision rule
Primary	Redemption rate	Share of assigned members with ≥ 1 completed redemption (any type) during the 12-week window.	Ship if lift is positive, significant ($\alpha = 0.05$), and $\geq$ the MDE on the pooled + post-stratified estimate.
Guardrail	Revenue per member (BRL)	Flown-flight revenue attributed to the member, window + 8 weeks post.	Non-inferiority: treatment not worse than control by > 3% (one-sided).
Guardrail	Point-liability cost per member (BRL)	Δ liability drawn down + cash cost of low-cost rewards redeemed.	Net effect reported with CI; flag if treatment cost per member is higher than control with 95% confidence.
Guardrail	Early-lapse / disengagement proxy	Share with no earn and no redemption in the 90 days after the window.	Directional; flag if treatment is worse.
Guardrail	Average redemption value & mix	Mean BRL value per redeeming member; share of redemptions that are low-cost-catalog only.	Context for whether the lift is "real engagement" or trivial-redemption gaming.

7.4 Power & minimum detectable effect

Assumption	Value
Control redemption rate over 12 weeks (planning value)	11.0%
Target (MDE), relative	+20% → absolute +2.2 pp (11.0% → 13.2%)
α (two-sided) / power	0.05 / 0.80

Assumption	Value
Required n per arm at the MDE (two-proportion)	≈ 3,450
Actual n per arm	≈ 7,500 → power > 0.95 at the MDE; ~ 0.80 power for an effect as small as +1.5 pp
Per-tier cells (per arm)	Blue 4,000 · Silver 2,000 · Gold 1,100 · Platinum 400 → well powered for Blue/Silver, directional only for Gold, underpowered for Platinum (a Platinum null = "consistent with no effect", not "proven zero")

The power calculation is re-run from the realized pre-period rates and reproduced in code in Phase 10; the numbers above are the planning basis.

7.5 Threats to validity the read-out must address

Threat	How it is handled
Sample-ratio mismatch	χ^2 goodness-of-fit on arm counts overall and per stratum; investigate if $p < 0.01$.
Covariate imbalance	Standardized mean differences on tier, tenure, pre-period earn, pre-period flights, pre-period balance, prior-redeemer flag; report a Love plot.
Novelty effect	Weekly treatment redemption rate across the 12 weeks; test for a decaying trend (weeks 1–4 vs 9–12); report the settled lift, not the peak.
Seasonality	Control arm is concurrent, so calendar seasonality differences out; still report both arms' weekly series.
Contamination / spillover	Household and referral links between arms assumed negligible; stated as a limitation with a rough bound.
Multiple comparisons	One primary metric fixed in advance; guardrails and the 4-way heterogeneity cut are treated as secondary and CI-reported, with a note on family-wise error.
Post-window horizon	Lapse is a 12-month construct; only 90-day forward proxies are available. Stated, not hidden.
Peeking	Single fixed analysis at window close; no interim looks drive the decision.

7.6 Heterogeneity analysis (required)

Effect on the primary metric estimated **within each tier**, with per-tier CIs and a formal **arm × tier interaction** test. The prior to test — not to assume — is that the effect is largest for **Blue** (threshold-constrained) and smallest for **Platinum** (already redeem freely). The business decision (full rollout vs rollout to lower tiers only) hinges on this cut.

8. The analytics window — why 24 months, and CY vs PY

Anchor	Value
Window length	24 complete months: month 1 = Sep 2024 , month 24 = Aug 2026
Current Year (CY)	months 13–24 → Sep 2025 – Aug 2026
Prior Year (PY)	months 1–12 → Sep 2024 – Aug 2025 — <i>the program's first, ramping year</i>
Experiment	months 19–21 → Mar–May 2026 (12 weeks)
Post-window observation	2026-05-25 → 2026-08-31 (~14 weeks)

Why 24 months. It is the full life of the relaunched program — earlier data is on the retired legacy scheme and not comparable. It is also exactly one SkyPoints lot lifetime, so breakage becomes observable within the window.

Comparability caveat. PY is the relaunch ramp: enrolment, earn and redemption were all climbing off a near-zero base, so **raw YoY overstates underlying growth**. Every YoY metric is reported with a **steady-state cut** alongside it — restricted to members enrolled **before month 4** (Dec 2024), whose behaviour is not dominated by onboarding. An account-level flag `is_steady_state_cohort` carries this.

9. Seasonality and patterns embedded in the data

Pattern	Where it shows
Jan / Jul / Dec Brazilian holiday travel peaks	Flight earn peaks; award-flight redemptions peak 1–2 months ahead (booking Oct–Nov for Dec–Jan, Apr–May for Jul)
Feb–Mar, May, Sep shoulder season	Lower earn; the experiment sits in this calmer stretch
Carnaval (Feb)	Short travel spike then a lull
Black Friday / Nov	Enrolment spike (card acquisition push), co-brand spend spike
Co-brand card earn	Steadier than flight earn; mild Nov–Dec holiday-spend lift
Point expiry wave	First large lot expiries in Jun–Aug 2026 (24 months after the relaunch surge) → breakage visibly steps up at window end
Tier re-evaluation	Monthly, but downgrades cluster ~12 months after the relaunch enrolment surge
Redemption into the experiment window	Naturally rising toward the Jul peak — handled by the concurrent control arm, not by de-seasonalising

10. Stakeholder personas → report pages

Four personas, one standing question each. Those four questions are the Phase-1 pages (dashboard is a secondary deliverable — kept lean).

#	Persona	Standing question	Page
1	Loyalty Director / CMO	Is the program healthy, and did Flash Redemption work well enough to roll out?	Executive Summary
2	Head of Loyalty Finance (liability owner)	Where is member value concentrated, how large and how fast-growing is the breakage liability, and how are members moving between tiers?	Member Tiers & Value
3	Experimentation / Product Lead	Is the effect real, significant and safe on the guardrails — or is it novelty, or one segment carrying it?	A/B Test Results
4	CRM / Retention Manager	Where do members fall out of the redeem journey, and who is about to lapse?	Redemption & Engagement Funnel

11. KPI framework

Benchmark ranges are **directional**, drawn from public airline-loyalty disclosures and industry commentary (IdeaWorks/Switchfly breakage studies, IATA loyalty notes), adapted to a young single-country program. They set targets; they are not claims about named competitors.

11.1 Program health & scale

KPI	Definition	Benchmark	Target
Enrolled members (cumulative)	Distinct accounts ever enrolled	—	on relaunch curve
Active members	Earn or flight in trailing 12m	—	≥ 80k by window end
Active rate %	Active ÷ enrolled	45%–65%	≥ 55%
New enrolments / month	First enrolment in the month	—	≥ plan
Member tenure mix	Share by <6 / 6–12 / 12–24 months	—	context
Co-brand card penetration %	Active members holding the card	12%–25%	≥ 20%

11.2 Earn & burn

KPI	Definition	Benchmark	Target
SkyPoints issued / month	Sum of earn events	—	≥ plan
SkyPoints redeemed / month	Sum of redemption debits	—	≥ plan

KPI	Definition	Benchmark	Target
Burn/earn ratio	Points redeemed ÷ points issued (trailing 12m)	0.55–0.80	≥ 0.65
Quarterly redemption rate %	Active members with ≥1 redemption in the quarter	8%–15%	≥ 15%
Ever-redeemed %	Active members with ≥1 redemption ever	30%–55%	≥ 45%
Avg SkyPoints balance (active)	Mean outstanding balance	—	context
Median months-to-first-redemption	From enrolment to first redemption	6–14	≤ 10

11.3 Breakage & liability

KPI	Definition	Benchmark	Target
Point liability (BRL)	Points outstanding × unit cost (gross and breakage-adjusted)	—	≤ plan
Breakage rate %	Cohort points issued that expire unredeemed	10%–30%	≤ 30%
Points expiring / month (BRL)	Value of lots reaching 24-month expiry	—	minimise
Liability per active member (BRL)	Liability ÷ active members	—	context
Liability coverage (months of redemptions)	Liability ÷ trailing-3m redemption value	—	context

11.4 Tiers & value

KPI	Definition	Benchmark	Target
Tier mix %	Active members by tier	~68/20/9/3	context
Upgrades / downgrades per month	Tier-change events by direction	—	net ≥ 0
Tier retention rate %	Members holding tier at re-evaluation	Silver+ 60%–80%	≥ 70%
Revenue per member by tier (BRL/yr)	Flown flight revenue ÷ members, by tier	rises steeply by tier	context
Share of flight revenue from Gold+Platinum	Top-tier revenue ÷ total member flight revenue	25%–45%	context

11.5 Retention & lapse

KPI	Definition	Benchmark	Target
12-month account lapse rate %	Active members who hit <i>account</i> lapse within 12 months	3%–8%	context
12-month engagement lapse rate %	Active members with no flight and no redemption for 12 months	15%–30%	≤ 22%
Active → lapsed / month	Monthly lapse count	—	minimise
Reactivation rate %	Lapsed members reactivating within 12m	8%–18%	≥ 12%
Redeemer lapse gap (pp)	Never-redeemer lapse rate – redeemer lapse rate (adjusted)	positive	quantify
Mn cohort retention	Share of an enrolment cohort still active at month n	M12 55%–75%	≥ 65%

11.6 Commercial value

KPI	Definition	Benchmark	Target
Revenue per active member (BRL/yr)	Flown flight revenue ÷ active members	—	≥ plan
Award-seat share %	Award-flight seats ÷ total seats	5%–12%	6%–10% (managed)
Points-plus-cash attach %	Bookings using SkyCash ÷ award bookings	—	context
Incremental flights from redeemers	Flights in 6m after first redemption vs matched non-redeemers	positive	quantify

11.7 Experiment scorecard (A/B Test Results page)

KPI	Definition
Redemption rate by arm (+ lift, CI, p)	Primary metric, \$7.3

KPI	Definition
Weekly redemption rate by arm	Novelty-effect diagnostic, \$7.5
SRM χ^2 (overall + per stratum)	Randomization check
Covariate balance (SMD / Love plot)	Randomization check
Revenue per member by arm (+ non-inferiority test)	Guardrail
Liability cost per member by arm (+ CI)	Guardrail
90-day post-window disengagement by arm	Guardrail
Redemption value & low-cost-only share by arm	Guardrail / context
Effect by tier (+ interaction test)	Heterogeneity, \$7.6

12. Targets and plan

- The plan is built on the **same basis as the actuals** (same lapse definition, same breakage-adjusted liability basis).
- **Plan active members:** relaunch business-case curve — ramp to **80k active** by month 24.
- **Plan breakage:** declining from ~40% at relaunch to **≤ 30%** by month 24.
- **Plan redemption rate:** rising to **≥ 15%** quarterly by month 24.
- **Plan 12-month lapse rate:** **≤ 22%**.
- **Plan liability:** issuance on the earn plan, redemptions on the burn plan → a planned liability trajectory; actual vs plan is the CFO scoreboard.
- "vs Target %" measures exist per KPI; **breakage rate, lapse rate and liability per member are scored lower-is-better**.

13. Multi-currency handling

Out of scope by design. All monetary values are **BRL**. No `*_local` / `*_usd` split, no FX table, no reporting-currency slicer. This choice is deliberate — it keeps the project's attention on the analysis and the experiment.

14. Out of scope (explicit)

- International routes, partner-airline accrual/redemption, Star/oneworld-style alliances.
- Fare pricing and revenue management (flight revenue is an input, not modelled).
- Fraud, points brokering, account-takeover.
- Crew, fuel, maintenance, aircraft economics — this is a **loyalty** project, not an airline-operations one.
- Real-time data; everything is monthly/weekly batch.
- A lapse-**prediction** model. Phase 10 quantifies drivers and the experiment effect; it does not ship a churn scorecard.
- Marketing-channel attribution beyond the experiment's own assignment.
- Real personal data. All members, flights, card spend, earn and redemption events are synthetic and seed-generated.
- The **true Flash Redemption effect size** — decided in Phase 3, not here.

15. Resolved-assumptions log

#	Decision	Resolution
A1	Window length & anchor	24 months, Sep 2024 – Aug 2026; data-gen date 2026-09-02; = relaunched-program life.
A2	Two point currencies	Spendable SkyPoints (24-month lot expiry, FIFO) + status Tier Points (reset on anniversary, non-redeemable).
A3	Tiers	Blue / Silver / Gold / Platinum; trailing-12m TP-or-segments qualification; 12-month tier tenure guarantee.
A4	Active member	Earn or flight in trailing 12 months.
A5	Lapsed member	12 consecutive months with no earn and no redemption.
A6	Breakage	Cohort-based estimate, aggregate roll-forward cross-check; program-to-date ~40%; plan ≤ 30%.

#	Decision	Resolution
A7	Unit cost for liability	≈ R\$0.021 / pt weighted; reported gross and breakage-adjusted; sensitivity table required.
A8	Experiment unit / split	Member; 50/50 within tier strata.
A9	Experiment eligibility	Active & not lapsed at 2026-03-01; frozen list ~58k.
A10	Experiment sample	15,000; stratified with Gold/Platinum oversampling (8,000 / 4,000 / 2,200 / 800).
A11	Primary metric	Redemption rate over the 12-week window.
A12	Guardrails	Revenue/member (non-inferiority –3%), liability cost/member, 90-day post disengagement, redemption value & mix.
A13	MDE / power	+20% relative (+2.2 pp on 11.0%); ~7,500/arm → >0.95 power at MDE.
A14	Required diagnostics	SRM, covariate balance, novelty decay, tier heterogeneity + interaction test.
A15	Post-window horizon	~14 weeks; 90-day proxies only for lapse; stated as a limitation.
A16	Currency	BRL only; no FX in scope.
A17	YoY comparability	PY = relaunch ramp; every YoY paired with a steady-state-cohort cut (<code>is_steady_state_cohort</code>).
A18	True treatment effect	Deliberately not decided here; set in ETL (Phase 3), detected blind in Phase 10.

End of Phase 1 deliverable.

AeroVanti SkyPoints — Data Architecture

Document status: Phase 2 deliverable. **Depends on:** 01_business_context_kpis.md . **Feeds:** Phase 3 (ETL build), Phase 4 (data dictionary / model docs), Phase 6 (semantic model), **Phase 10 (analysis — primary deliverable).**

This is a build spec: every layer, every table, every data-quality issue Phase 3 must inject and then handle. Scope note from Phase 0: the messy-data layer is **reduced** — 6 simulated source systems, 6 primary data-quality issue classes. Effort is concentrated on a curated layer that serves the A/B test read-out cleanly.

1. Principles — medallion, seeded, idempotent

Data only ever flows downward.

Layer	Folder	Contract
Raw	data/raw/<source>/	Byte-faithful simulated exports — messy, locale-specific, duplicated, encoded however the fake source system would encode it. Never edited after generation. Regenerated only when SEED / config.py changes.
Staging	data/staging/	Exactly one typed table per raw entity: stg_<source>_<entity>.parquet . Parsing, type coercion, encoding repair, de-duplication, key & category normalisation. No cross-source joins, no business logic, no metric definitions. Every fix counted into a DQ fragment.
Curated	data/curated/	Conformed Kimball star: dim_* / fact_* as Parquet and a CSV mirror. Surrogate keys, tier-status-from-ledger, FIFO point-lot accounting, breakage cohorting, the pre-built experiment outcome tables, is_steady_state_cohort . All money in BRL (no FX).
Quality	data/quality/	DQ fragments from staging + curated, assembled into dq_report.md ; hard-assertion results from dq_checks.py .

Seeded. A single SEED in etl/config.py drives every random draw. Same seed → byte-identical raw → identical curated. **The true Flash Redemption effect lives only in config.py** (see §10).

Idempotent. Each script wipes and rebuilds its own layer. run_pipeline.py runs 01 → 02 → 03 → dq_checks end to end; dq_checks.py exits non-zero on any violated assertion.

Layered. Staging reads only raw. Curated reads only staging. The semantic model (Phase 6) and the analysis (Phase 10) read only data/curated/*.parquet .

2. Tech stack & run instructions

Concern	Choice
Language	Python 3.11+
Core libs	pandas , numpy , pyarrow (Parquet), python-dateutil , openpyxl (write messy .xlsx), Faker (pt_BR locale — names, cities, CPF-like ids), ftfy + unicodedata (encoding repair in staging), duckdb (Phase 10 SQL track + optional staging SQL), scipy / statsmodels (Phase 10 tests — not used in ETL)
Storage	Flat files. Raw = .csv / .xlsx / .csv.gz / .parquet . Staging & curated = .parquet (+ curated .csv mirror). No database server.
Orchestration	Plain scripts, ordered by run_pipeline.py . No Airflow / dbt.
Determinism	One SEED ; every generator derives a named sub-seed from it.
Windows / OneDrive	Write to a temp path in the same folder, then atomic os.replace ; a retry-on-lock wrapper (utils.write_parquet , ~4 attempts, backoff) absorbs intermittent WinError 5/32 locks.

```
# from repo root
python -m pip install -r etl/requirements.txt
python etl/run_pipeline.py          # full build: raw -> staging -> curated -> dq_checks
```

```
python etl/run_pipeline.py --from 02 # rebuild staging + curated only
python etl/gen_data_dictionary.py      # regenerate docs/03_data_dictionary.md from disk
```

Run order inside `run_pipeline.py` :

1. `01_generate_raw.py` — writes `data/raw/**`
2. `02_clean_stage.py` — writes `data/staging/stg_*.parquet` + `data/quality/dq_fragments/stg_*.json`
3. `03_build_curated.py` — writes `data/curated/{dim,fact}_*.parquet, csv` + curated DQ fragments
4. `dq_checks.py` — hard assertions; assembles `data/quality/dq_report.md` ; non-zero exit on failure

3. Naming conventions

Thing	Rule	Example
Raw file	<code><source>__<entity>__<period-or-grain>.<ext></code>	<code>skycore__point_transactions__2025-07.csv</code>
Staging table	<code>stg_<source>_<entity></code>	<code>stg_skycore_point_transactions</code>
Dimension	<code>dim_<noun></code>	<code>dim_member</code>
Fact	<code>fact_<process>_<grain></code>	<code>fact_point_transaction</code>
Surrogate key	<code><entity>_key</code> — integer, curated only	<code>member_key</code>
Business key	<code><entity>_id</code> — preserved from source	<code>member_id</code>
Date	<code><role>_date (date) · <role>_date_key (int YYYYMMDD)</code>	<code>enrollment_date</code> , <code>enrollment_date_key</code>
Timestamp	<code><role>_ts</code> — tz-aware, America/Sao_Paulo in curated	<code>assigned_ts</code>
Money	<code><measure>_brl</code> (single currency)	<code>flight_revenue_brl</code> , <code>liability_brl</code>
Points	<code><measure>_pts</code> or <code>points_<x></code>	<code>points_earned_m</code> , <code>balance_pts</code>
Boolean	<code>is_<x></code> / <code>has_<x></code>	<code>is_lapsed_eom</code> , <code>has_cobrand_card</code>
Columns	<code>snake_case</code> throughout curated	

Source timestamps are parsed to tz-aware UTC in staging, then curated derives the **Brazil-local calendar day** (`America/Sao_Paulo`) and every `*_date_key` from that local day, so month-boundary and experiment-week-boundary events land in the right bucket.

4. Source-system inventory

Six simulated systems. "Export style" is what makes each one messy.

#	System (fictional)	Owns	Raw entities	Export style / quirks
1	SkyCore Loyalty Platform	Program of record: members, point ledger, tier snapshot and tier-change events, reward catalog	<code>members_snapshot</code> (daily full dump), <code>point_transactions</code> (monthly), <code>tier_snapshots_monthly</code> , <code>tier_change_events</code> , <code>reward_catalog</code>	Daily full-dump member snapshot → massive row duplication ; <code>point_transactions</code> split into monthly files with overlapping date ranges → cross-file dupes; <code>points</code> and any BRL value occasionally exported as text; <code>member_since</code> mixes <code>YYYY-MM-DD</code> and <code>DD/MM/YYYY</code> ; <code>tier_snapshots_monthly</code> drifts from <code>tier_change_events</code> (snapshot job runs on stale replica).
2	BancoAV Co-Brand Card Feed	Co-branded Mastercard: card accounts, monthly spend, SkyPoints issued from spend	<code>card_accounts</code> , <code>card_spend_monthly</code>	Bank export in Latin-1 → mojibake on holder names / city; <code>valor_fatura</code> as "R\$ 1.234,56" ; <code>competencia</code> (statement month) as Excel serial ; the June 2025 file was delivered twice with a different filename → duplicate month.

#	System (fictional)	Owns	Raw entities	Export style / quirks
3	Reserva Booking System	Flights: bookings, flown segments, base fare, fare family, award vs cash	bookings , segments	Third-party PSS export with <code>MM/DD/YYYY</code> dates; taxes sometimes inside <code>fare_total</code> , sometimes separate; <code>loyalty_member_id</code> blank for non-member bookings and, for ~2% of member bookings, points to a <code>member_id</code> absent from SkyCore (guest→member merge lag); duplicate segment rows on schedule changes.
4	Aurora CRM	Member contact attributes, lifecycle stage, campaign membership, contact consent	crm_contacts , campaign_membership	Free-text <code>lifecycle_stage</code> ; <code>data_nascimento</code> sometimes <code>1900-01-01</code> or <code>0</code> ; CRM went live in program month 4 , so members enrolled in months 1–3 have no CRM row ; <code>cidade</code> with encoding damage; overlapping campaign memberships.
5	Flesk Experiment Platform	A/B assignment for Flash Redemption: arm, stratum, enrolment, feature-flag exposure	ab_assignments , exposure_log	Modern tool, mostly clean — but <code>exposure_log</code> is at-least-once (duplicate rows); all timestamps UTC with Z while every other system is Brazil-local; ~30 <code>ab_assignments</code> rows are late eligibility corrections for members who did not meet the frozen 2026-03-01 criteria and must be reconciled out.
6	Loyalty Liability Sub-Ledger (Finance)	Monthly points issued/redeemed/expired totals, liability balance, breakage provision, cash cost of partner rewards	liability_rollforward_monthly , reward_cost_monthly	<code>.xlsx</code> , one row per month; free-text account labels; amounts as text with parentheses negatives "(1.234,00)" ; <code>mes</code> column as Excel serial ; months 2025-11 and 2026-02 were restated (two versions in the file, later one wins).

`tier_change_events` (SkyCore) is the **movement source of truth**; `tier_snapshots_monthly` is kept only to *measure* drift (DQ issue Q06). Flesk (#5) corresponds to the experiment itself — a designed intervention, not "neutral history".

5. Raw layer — what `01_generate_raw.py` emits

Volumes are approximate at `SEED` default.

Raw file(s)	Grain	~Rows	Key messy traits injected
<code>skycore__members_snapshot_YYYY-MM-DD.csv</code> (daily)	member × snapshot day	~40M (dupe-heavy)	full re-dump daily; <code>member_since</code> mixed date formats; <code>home_city</code> casing noise
<code>skycore__point_transactions_YYYY-MM.csv</code>	one ledger entry	~3.3M	overlapping monthly ranges → cross-file dupes; <code>points</code> / <code>value_br1</code> sometimes text; <code>txn_type</code> free text (<code>earn</code> / <code>Earn</code> / <code>ACCRUAL</code> , <code>redeem</code> / <code>redemption</code> , <code>expire</code> / <code>expiry</code>)
<code>skycore__tier_snapshots_monthly.csv</code>	member × month	~2.4M	drifts from the event ledger (~3–6% of member-months disagree)
<code>skycore__tier_change_events.csv</code>	tier-change event	~250k	the movement truth; a few duplicate <code>event_id</code> (at-least-once)
<code>skycore__reward_catalog.xlsx</code>	reward option × effective period	~40	Excel-serial <code>available_from</code> ; <code>points_price</code> as text; the Flash low-cost rows have <code>available_from</code> = <code>2026-03-02</code>
<code>bancoav__card_accounts.csv</code>	card account	~34k	Latin-1 names/city; <code>abertura</code> date as <code>DD/MM/YYYY</code> ; some accounts with no matching SkyCore member (pre-enrolment applications)

Raw file(s)	Grain	~Rows	Key messy traits injected
bancoav__card_spend_monthly.csv	card account × month	~700k	valor_fatura as "R\$ 1.234,56" ; competencia Excel serial; June 2025 duplicated
reserva__bookings.csv	booking (PNR)	~1.4M	MM/DD/YYYY dates; channel free text
reserva__segments.csv	flown segment	~1.9M	taxes bundled/unbundled inconsistently; blank / orphan loyalty_member_id ; duplicate rows on schedule change
aurora__crm_contacts.csv	member	~135k	free-text lifecycle stage; data_nascimento 1900/0; missing months-1–3 cohorts; cidade mojibake
aurora__campaign_membership.csv	member × campaign	~180k	campaign-name noise; overlapping memberships
flesk__ab_assignments.csv	assigned member	~15,030	~30 ineligible late-correction rows to reconcile out; stratum present
flesk__exposure_log.csv	member × exposure event	~120k	at-least-once duplicates; UTC Z timestamps
ledger__liability_rollforward_monthly.xlsx	month	~26	parentheses negatives; Excel-serial month; 2 restated months (dupe rows)
ledger__reward_cost_monthly.xlsx	month × reward category	~150	text amounts; free-text category labels

6. Staging layer — 02_clean_stage.py

One `stg_*` table per raw entity. Responsibilities and the DQ counter each raises:

Fix class	What happens	DQ counter(s)
Date parsing	Parse YYYY-MM-DD , DD/MM/YYYY , MM/DD/YYYY , ISO-8601 with TZ, and Excel serials (epoch 1899-12-30). Ambiguous 01/02/2025 resolved by source-locale rule (Reserva = US, everything else = BR).	dates_parsed , excel_serials_converted , dates_unparseable
Money parsing	Strip R\$, thousands . and decimal , ; " (x)" → -x ; blank → null (never 0).	money_strings_parsed , money_negatives_from_parens
Points parsing	Coerce points to int; sign by txn_type (earn +, redeem/expire -); text magnitudes ("1.500") parsed.	points_strings_parsed , points_sign_normalised
Encoding repair	ftfy.fix_text + targeted Latin-1→UTF-8 re-decode for BancoAV / Aurora; strip BOM.	mojibake_repaired , bom_stripped
De-duplication	Drop exact-duplicate event_id / txn_id / exposure rows; collapse daily member full-dump to one row per member per day; dedupe overlapping point_transactions monthly files on txn_id ; drop the re-delivered BancoAV June-2025 file; keep only the latest restatement of a ledger month.	dupe_events_dropped , snapshot_rows_collapsed , dupe_txn_dropped , restated_months_resolved
Timezone	All timestamps → tz-aware UTC (Flesk already Z ; others localised from America/Sao_Paulo).	timestamps_localized
Category harmonisation	Map txn_type , lifecycle_stage , booking channel , tier labels, reward categories to controlled vocab in etl/vocab/ . Unknowns → explicit "unknown" + logged.	categories_normalised , category_unknowns
Key normalisation	Trim business keys; cast to string; null-out placeholders ("0" , "" , "N/A"). Blank loyalty_member_id on bookings → explicit null (non-member).	null_keys_flagged
Birth-year sanity	data_nascimento ∈ {1900-01-01, 0, future} → null + birth_year_valid = false .	birthyear_invalidated

Fix class	What happens	DQ counter(s)
Type coercion	Enforce a declared dtype per column; failures → null + counter.	type_coercion_failures

Staging does **not** join across sources, does **not** compute balances / lapse / breakage, and does **not** drop rows for referential problems — that is curated's job, fully accounted.

7. Curated layer — 03_build_curated.py

Kimball star. `dim_date` is built first and spans **2024-01-01 → 2027-12-31** (well past the fact window, per the guardrail about a stray "(blank)" date member when point lots expire and tiers re-evaluate past window end).

7.1 Dimensions

Table	Grain	Notable attributes
<code>dim_date</code>	calendar day	<code>date_key</code> , y/q/m/iso-week, <code>month_name</code> , <code>is_month_end</code> , <code>is_holiday_br</code> , <code>season_tag</code> , <code>experiment_phase</code> (pre / test / post), <code>experiment_week</code> (1–12 during 2026-03-02→2026-05-24, else null), hidden sort keys
<code>dim_member</code>	member	<code>member_key</code> , <code>member_id</code> , <code>enrollment_date_key</code> , <code>enrollment_cohort_month</code> , <code>enrollment_channel</code> (flight / card / web / partner), <code>home_city</code> , <code>home_region</code> (BR macro-region), <code>has_cobrand_card</code> (ever), <code>current_tier_key</code> , <code>age_band</code> , <code>birth_year_valid</code> , <code>crm_lifecycle_stage</code> , <code>is_steady_state_cohort</code> (enrolled ≤ 2024-12), <code>is_test_account</code> , <code>experiment_arm</code> (control / treatment / not_enrolled), <code>experiment_stratum</code> (tier at assignment, else n/a), <code>is_experiment_subject</code>
<code>dim_tier</code>	tier	<code>tier_key</code> , <code>tier_name</code> (Blue/Silver/Gold/Platinum), <code>tier_rank</code> (1–4), <code>earn_multiplier</code> , <code>tp_threshold</code> , <code>segment_threshold</code> , <code>is_elite</code> (Silver+)
<code>dim_earn_source</code>	earn source	<code>earn_source_key</code> , <code>source_name</code> (flight / cobrand_card / partner_hotel / partner_car / partner_retail / promo / service_recovery), <code>source_group</code> (flight / card / partner / other), <code>earns_tier_points</code>
<code>dim_reward_type</code>	redemption option	<code>reward_type_key</code> , <code>reward_name</code> , <code>reward_category</code> (award_flight / seat / bag / lounge / gift_card / voucher), <code>points_price_typical</code> , <code>value_per_point_br1</code> , <code>is_low_cost_catalog</code> (the Flash additions), <code>available_from_date_key</code>
<code>dim_route</code>	route	<code>route_key</code> , <code>origin</code> , <code>dest</code> , <code>haul_band</code> (short / medium), <code>region_pair</code> — slim; flights are not a deep thread
<code>dim_experiment_arm</code> (disconnected slicer helper, built Phase 6)	arm	control / treatment / not_enrolled
<code>dim_experiment_stratum</code> (disconnected slicer helper)	stratum	Blue / Silver / Gold / Platinum

7.2 Facts

Table	Grain	Measures / columns	Build notes
<code>fact_point_transaction</code>	one ledger entry	<code>member_key</code> , <code>txn_date_key</code> , <code>txn_type</code> (earn / redeem / expire / adjust), <code>earn_source_key</code> , <code>reward_type_key</code> , <code>points</code> (signed), <code>points_value_br1</code> , <code>earn_month_key</code> (issuing lot), <code>lot_id</code> , <code>booking_id</code> , <code>is_experiment_window</code>	Deduped, ordered. FIFO lot linkage: each redeem/expire row consumes named lots; <code>earn_month_key</code> carries the issuing cohort for breakage attribution. Spine of earn/burn/breakage.
<code>fact_member_month</code>	member × month	<code>is_active_eom</code> , <code>is_lapsed_eom</code> , <code>became_lapsed_this_month</code> , <code>is_reactivation</code> , <code>months_since_last_activity</code> , <code>tier_key</code> (ledger-derived), <code>balance_pts_eom</code> , <code>points_earned_m</code> , <code>points_redeemed_m</code> , <code>points_expired_m</code> , <code>redemptions_m</code> , <code>has_redeemed_ever_eom</code> , <code>flights_m</code> ,	The retention / program-health workhorse. Lapse = 12 consecutive months with no earn and no redemption; for the first 11 months from window start <code>is_lapsed_eom</code> is null (undeterminable) — not <code>false</code> . Once lapsed, stays lapsed until a reactivation row.

Table	Grain	Measures / columns	Build notes
		flight_revenue_brl_m , card_spend_brl_m , tenure_months , is_new_enrollment	
fact_flight_segment	one flown segment	member_key (sentinel -1 non-member / -2 orphan), flight_date_key , booking_date_key , route_key , fare_family , base_fare_brl , taxes_brl , booking_channel , is_award_redemption , points_redeemed , skypoints_earned , tier_points_earned	Taxes reconciled to a single definition; duplicate schedule-change rows collapsed. Supports award-seat share, revenue per member, flight earn.
fact_card_spend_month	member × month	card_spend_brl , skypoints_earned , is_card_active	From BancoAV; joined to member on card→member map.
fact_tier_change	one tier-change event	member_key , change_date_key , from_tier_key , to_tier_key , direction (upgrade / downgrade / retain_drop), trigger (qualification / soft_landing / manual)	Ledger-derived, deduped. Curated tier status is folded forward from this , never from the snapshot.
fact_liability_month	month	points_outstanding_eop , liability_brl_gross , liability_brl_breakage_adj , points_issued_m , points_redeemed_m , points_expired_m , breakage_provision_brl , reward_cash_cost_brl	From the sub-ledger, reconciled to fact_point_transaction roll-up within 0.5% (DQ check). Unit cost per doc 01 \$5.3.
fact_experiment_assignment	one assigned member (~15,000)	member_key , arm_key , stratum , assigned_ts , pre_earn_12m_pts , pre_flights_12m , pre_redemptions_12m , pre_balance_pts , prior_redeemer_flag , pre_tenure_months	Ineligible late-correction rows reconciled out. Pre-period covariates frozen at 2026-03-01 — used for the SRM / balance checks and CUPED-style variance reduction in Phase 10.
fact_experiment_member_week	assigned member × experiment week (~180,000)	arm_key , redeemed_w , redemptions_w , points_redeemed_w , redemption_value_brl_w , low_cost_only_w , earn_w , flights_w , flight_revenue_brl_w	The weekly series for the novelty-effect diagnostic.
fact_experiment_outcome	one assigned member (~15,000)	redeemed_in_window (primary), redemptions_in_window , redemption_value_brl , low_cost_only , revenue_brl_window_plus8w , liability_drawdown_brl , reward_cash_cost_brl , net_liability_cost_brl , disengaged_90d_post , earn_in_window , flights_in_window , active_post_quarter	One tidy row per subject with every metric the read-out needs , pre-computed so Phase 10 goes straight to tests and the report needs almost no DAX. redeemed_in_window == OR of the weekly flags (DQ check).
fact_targets_month	month × kpi	target_value	Plan from doc 01 \$12; same basis as actuals.

7.3 Cross-cutting curated rules

- **Tier status from ledger.** fact_member_month.tier_key and dim_member.current_tier_key are folded forward from fact_tier_change . tier_snapshots_monthly is used **only** to compute a drift metric (member-months where snapshot ≠ ledger-derived), reported in the DQ report, never as input.
- **FIFO point-lot accounting.** Every redeem / expire consumes the oldest open lots; earn_month_key on each consuming row ties breakage and redemption back to the **issuance cohort**, so cohort-based breakage (doc 01 \$5.3) is computable.
- **Breakage-adjusted liability.** liability_brl_breakage_adj = points outstanding × R\$0.021 × (1 – expected breakage), with expected_breakage a config.py parameter; the gross figure is also kept. A sensitivity table over the breakage assumption is a Phase-10 artifact, not baked here.

- **Lapse honesty.** `is_lapsed_eom` is null for the first 11 months and for any member with < 12 months tenure — the 12-month rule is simply not yet evaluable. Never coerced to false.
- **Sentinel keys.** `member_key` = -1 (non-member booking), -2 (orphan — member id not in master). Facts are **never dropped** for a missing dimension; the sentinel is used and counted.
- **Experiment denormalisation.** `experiment_arm` / `experiment_stratum` / `is_experiment_subject` are stamped on `dim_member` and propagated to `fact_member_month` via `member_key`, so the report can slice program health by arm without touching the experiment facts.
- **Local calendar day** (`America/Sao_Paulo`) drives every `*_date_key`, including the experiment-week bucketing.
- **BRL only.** Every money column is `*_br1`. No `_usd` sibling, no FX table.

8. Data-quality issue catalogue (the injection list)

Six **primary** classes — injected by `01`, handled by `02` / `03`, counted into a DQ fragment, and (where an invariant) asserted in `dq_checks.py`.

#	Issue	Injected in	Handled in	Hard check
Q01	Multiple date formats + Excel serials	skycore <code>member_since</code> , bancoav, <code>reserva</code> (<code>MM/DD/YYYY</code>), ledger, <code>reward_catalog</code>	staging date parser (locale rule per source)	<code>dates_unparseable = 0</code>
Q02	Money as locale strings / parentheses negatives / symbol-in-value	bancoav <code>card_spend_monthly</code> , ledger <code>.xlsx</code> , some skycore <code>value_br1</code>	staging money parser	no non-null money column is object dtype; <code>liability_br1_gross ≥ 0</code>
Q03	Encoding damage (Latin-1 mojibake, BOM)	bancoav names/city, aurora <code>cidade</code>	<code>ftfy</code> + targeted re-decode	no <code>Ã / Â</code> mojibake sequences remain in <code>dim_member.home_city</code>
Q04	Duplicate rows — daily full-dump snapshot, overlapping monthly <code>point_transactions</code> , re-delivered BancoAV month, at-least-once event/exposure rows	skycore, bancoav, flesk	staging de-dup (collapse / dedupe on id / drop re- delivered file)	<code>fact_point_transaction</code> unique on <code>txn_id</code> ; <code>fact_flight_segment</code> unique on <code>segment_id</code> ; snapshot collapsed to 1 row / member / day
Q05	Missing master rows / orphan keys — bookings referencing members absent from SkyCore, point rows for unknown members, months-1–3 cohorts with no CRM row	<code>reserva</code> , skycore, aurora late go-live	curated sentinel keys + count; CRM attributes left null with a <code>crm_present</code> flag	every fact FK resolves to a real or sentinel key; orphan count reported
Q06	Snapshot-vs-ledger drift — <code>tier_snapshots_monthly</code> disagrees with <code>tier_change_events</code>	skycore	curated builds status from the event ledger; drift measured	curated tier status == ledger-derived by construction ; drift-row count reported (expected 3–6% of member- months)

Also handled, without ceremony (minor, single counter each, no dedicated narrative): null/placeholder keys → sentinels; invalid birth years → null + flag; negative or out-of-range points/values → clamped + counted; Flesk UTC vs Brazil-local timezone bleed → converted; ledger restated months → latest wins; `txn_type` / channel / lifecycle free-text → controlled vocab.

`dq_checks.py` also asserts the **accounting / roll-up identities**:

- **Points roll-forward**, per member per month **and** program-wide per month: `opening_balance + earned - redeemed - expired = closing_balance` (exact, integer points).
- **Liability tie-out:** `fact_liability_month` issued / redeemed / expired totals match the `fact_point_transaction` roll-up within 0.5%; closing liability = opening + issued – redeemed – expired – revaluation.
- **Experiment integrity:** every `fact_experiment_member_week` / `fact_experiment_outcome` `member_key` ∈ `fact_experiment_assignment`; `redeemed_in_window` == OR of weekly flags; allocation ratio ∈ [0.49, 0.51] within every stratum (a *build-time* sanity check — the real SRM test is on observed data in Phase 10); pre-period covariates present for all subjects; data extends ≥ 90 days past 2026-05-24.
- **Retention consistency:** a lapsed member stays lapsed until a reactivation row; cohort retention curves are monotone non-increasing and ≤ 100%.
- **Benchmark bands:** redemption rate, breakage rate, 12-month lapse rate, burn/earn ratio and tier mix all land inside their doc 01 §11 ranges for the **pre-experiment steady-state** period (the experiment arms are allowed to diverge — that is the

point).

9. Architecture diagram

```
flowchart TD
    subgraph SRC["6 simulated source systems"]
        S1["SkyCore Loyalty Platform<br/>members - point ledger - tier snapshot + events - catalog"]
        S2["BancoAV Co-Brand Card Feed<br/>card accounts - monthly spend - card earn"]
        S3["Reserva Booking System<br/>bookings - flown segments - award vs cash"]
        S4["Aurora CRM<br/>contact attributes - lifecycle - campaigns"]
        S5["Flesk Experiment Platform<br/>A/B assignment - arm - stratum - exposure log"]
        S6["Loyalty Liability Sub-Ledger<br/>monthly liability roll-forward - reward cash cost"]
    end

    SRC -->| "01_generate_raw.py<br/>seeded, messy exports" | RAW["data/raw/**<br/>CSV - XLSX - CSV.GZ"]
    RAW -->| "02_clean_stage.py<br/>parse - repair - de-dup - normalise<br/>+ DQ fragments" | STG["data/staging/<br/>"]
    STG -->| "03_build_curated.py<br/>conform - surrogate keys - tier-from-ledger<br/>FIFO point lots - breakage co" | DQF["data/quality/dq_fragments/"]
    DQF -->| "dq_checks.py<br/>hard assertions + accounting identities" | DQR["data/quality/dq_report.md<br/>(build"]

    CUR -->| "Phase 6 - gen_semantic_model.py" | SM["Power BI .SemanticModel<br/>TMDL + Parquet import partitions"]
    SM -->| "Phase 8 - build_report.sh (lite)" | RPT["Power BI .Report<br/>4 pages"]
    CUR ==>| "Phase 10 - DuckDB (SQL) + pandas (Python)<br/>PRIMARY DELIVERABLE" | ANA["data_analysis/<br/>program d
```

10. Open items carried into Phase 3

#	Item	Resolution plan
B1	The true Flash Redemption effect — overall lift, per-tier heterogeneity profile, novelty-decay shape, and the guardrail impacts (revenue/member, liability cost/member, 90-day disengagement)	Decided in Phase 3 , encoded in <code>config.py</code> as tunable effect sizes + noise; kept out of docs 00/01; detected blind in Phase 10. Design intent: a real, non-trivial but not-obvious overall lift; effect concentrated in lower tiers; a modest early novelty bump that decays; guardrails roughly neutral with at least one that needs a careful (non-inferiority / borderline) read.
B2	Final row volumes vs the ~3M point-transaction / ~2.4M member-month targets	Calibrate <code>SEED</code> -default population and activity rates in the first regeneration pass.
B3	Benchmark-band calibration (redemption rate, breakage, burn/earn, lapse, tier mix, months-to-first-redemption, award-seat share)	Several regeneration passes until the pre-experiment steady-state sits mid-band for every doc 01 §11 KPI.
B4	BR holiday / school-break calendar & travel seasonality curve	Static lookup in <code>etl/vocab/br_calendar.csv</code> ; seasonal multipliers in <code>config.py</code> .
B5	Co-brand card penetration curve and card-earn share of issuance	Seeded logistic adoption curve; target ~20% active-member penetration, ~42% of issuance, at window end.
B6	Exact eligible-pool size and per-stratum counts	Generator targets ~58k eligible; assignment draws exactly 8,000 / 4,000 / 2,200 / 800, 50/50 within stratum.

End of Phase 2 deliverable.

AeroVanti SkyPoints — Data Dictionary

Auto-generated from `data/curated/` on 2026-09-02 · SEED=20260902. Do not edit by hand — run `python etl/gen_data_dictionary.py`.

18 curated tables.

`dim_date` • 1,461 rows

Calendar spine, 2024-01-01 to 2027-12-31. Carries the experiment phase/week tags and BR holiday / season attributes.

column	dtype	non-null	distinct	example
<code>date</code>	object	100.0%	1,461	2024-01-01 00:00:00
<code>date_key</code>	object	100.0%	1,461	20240101
<code>year</code>	object	100.0%	4	2024
<code>quarter</code>	object	100.0%	4	1
<code>month</code>	object	100.0%	12	1
<code>month_key</code>	object	100.0%	48	202401
<code>month_name</code>	object	100.0%	48	Jan 2024
<code>month_sort</code>	object	100.0%	48	24289
<code>iso_week</code>	object	100.0%	53	1
<code>day_of_week</code>	object	100.0%	7	0
<code>is_month_end</code>	object	100.0%	2	False
<code>is_school_break</code>	object	100.0%	2	False
<code>is_holiday_br</code>	object	100.0%	2	False
<code>season_tag</code>	object	100.0%	4	Summer peak
<code>experiment_phase</code>	object	100.0%	4	out_of_window
<code>experiment_week</code>	object	5.7%	12	1

`dim_earn_source` • 9 rows

SkyPoints issuance sources; -1 = unknown.

column	dtype	non-null	distinct	example
<code>earn_source_key</code>	object	100.0%	9	-1
<code>source_name</code>	object	100.0%	9	unknown
<code>source_group</code>	object	100.0%	4	other
<code>earns_tier_points</code>	object	100.0%	2	False

`dim_experiment_arm` • 3 rows

Slicer helper: control / treatment / not_enrolled.

column	dtype	non-null	distinct	example
<code>arm_key</code>	object	100.0%	3	control

dim_experiment_stratum • 4 rows

Slicer helper: the tier strata used for randomisation.

column	dtype	non-null	distinct	example
stratum	object	100.0%	4	Blue

dim_member • 120,000 rows

One row per SkyPoints member. Enrolment attributes + current ledger-derived tier + experiment arm/stratum.

column	dtype	non-null	distinct	example
member_key	object	100.0%	120,000	1
member_id	object	100.0%	120,000	SKP100000
enrollment_date_key	object	100.0%	648	20250714
enrollment_cohort_month	object	100.0%	24	2025-07
enrollment_channel	object	100.0%	4	card
home_city	object	100.0%	21	Vitória
home_region	object	100.0%	5	Sudeste
has_cobrand_card	object	100.0%	2	True
current_tier_key	object	100.0%	4	1
current_tier_name	object	100.0%	4	Blue
crm_present	object	100.0%	2	True
crm_lifecycle_stage	object	100.0%	5	engaged
age_band	object	63.3%	5	35-44
is_steady_state_cohort	object	100.0%	2	False
is_test_account	object	100.0%	1	False
experiment_arm	object	100.0%	3	not_enrolled
experiment_stratum	object	100.0%	5	n/a
is_experiment_subject	object	100.0%	2	False

dim_reward_type • 10 rows

Redemption catalogue; is_low_cost_catalog flags the Flash additions; -1 = unknown.

column	dtype	non-null	distinct	example
reward_type_key	object	100.0%	10	-1
reward_name	object	100.0%	10	unknown
reward_category	object	100.0%	7	unknown
points_price	object	90.0%	9	8000.0
value_per_point_brl	object	90.0%	9	0.024
is_low_cost_catalog	object	100.0%	2	False
available_from_date_key	object	90.0%	2	20240901.0
available_from	object	90.0%	2	2024-09-01 00:00:00

dim_route • 290 rows

Distinct flown routes with a haul band; -1 = unknown.

column	dtype	non-null	distinct	example
route_key	object	100.0%	290	-1
origin	object	100.0%	18	?
dest	object	100.0%	18	?
haul_band	object	100.0%	3	unknown
region_pair	object	100.0%	290	?

dim_tier • 4 rows

The four tiers (Blue/Silver/Gold/Platinum) with earn multipliers and thresholds.

column	dtype	non-null	distinct	example
tier_key	object	100.0%	4	1
tier_name	object	100.0%	4	Blue
tier_rank	object	100.0%	4	1
earn_multiplier	object	100.0%	4	1.0
tp_threshold	object	100.0%	4	0
segment_threshold	object	100.0%	4	0
is_elite	object	100.0%	2	False

fact_card_spend_month • 670,216 rows

Member x month co-brand card spend and card SkyPoints earned.

column	dtype	non-null	distinct	example
member_key	object	100.0%	46,881	1
month_key	object	100.0%	24	202507
card_spend_br1	object	100.0%	281,654	1780.85
skypoints_earned	object	100.0%	4,469	712
is_card_active	object	100.0%	1	True

fact_experiment_assignment • 15,000 rows

One row per assigned subject (eligible only; late corrections reconciled out). Frozen pre-period covariates.

column	dtype	non-null	distinct	example
member_key	object	100.0%	15,000	16
member_id	object	100.0%	15,000	SKP100015
arm_key	object	100.0%	2	treatment
stratum	object	100.0%	4	Gold
assigned_ts	object	100.0%	13,749	2026-03-01 09:21:14+00:00
pre_earn_12m_pts	object	100.0%	10,408	35061
pre_flights_12m	object	100.0%	71	20

column	dtype	non-null	distinct	example
pre_redemptions_12m	object	100.0%	8	0
pre_balance_pts	object	100.0%	11,078	36260
prior_redeemer_flag	object	100.0%	2	False
pre_tenure_months	object	100.0%	18	14

fact_experiment_member_week • 180,000 rows

Assigned subject x experiment week — the weekly redemption series for the novelty diagnostic.

column	dtype	non-null	distinct	example
member_key	object	100.0%	15,000	16
member_id	object	100.0%	15,000	SKP100015
arm_key	object	100.0%	2	treatment
stratum	object	100.0%	4	Gold
week	object	100.0%	12	1
redeemed_w	object	100.0%	2	False
redemptions_w	object	100.0%	2	0
points_redeemed_w	object	100.0%	534	0
redemption_value_brl_w	object	100.0%	571	0.0
low_cost_only_w	object	100.0%	2	False

fact_experiment_outcome • 15,000 rows

One row per subject with the full window + post-window outcomes: primary metric, guardrails, diagnostics.

column	dtype	non-null	distinct	example
member_key	object	100.0%	15,000	16
member_id	object	100.0%	15,000	SKP100015
arm_key	object	100.0%	2	treatment
stratum	object	100.0%	4	Gold
redeemed_in_window	object	100.0%	2	False
redemptions_in_window	object	100.0%	4	0
redemption_value_brl	object	100.0%	688	0.0
points_redeemed_in_window	object	100.0%	588	0
low_cost_only	object	100.0%	2	False
revenue_brl_window_plus8w	object	100.0%	9,329	4890.93
liability_drawdown_brl	object	100.0%	588	0.0
reward_cash_cost_brl	object	100.0%	566	0.0
net_liability_cost_brl	object	100.0%	633	0.0
disengaged_90d_post	object	100.0%	2	False
earn_in_window	object	100.0%	2	True
flights_in_window	object	100.0%	24	8

column	dtype	non-null	distinct	example
active_post_quarter	object	100.0%	2	True

fact_flight_segment • 672,320 rows

One flown segment. member_key -1 = non-member booking, -2 = orphan. skypoints_earned recomputed with the month's tier multiplier.

column	dtype	non-null	distinct	example
member_key	object	100.0%	89,251	16657
loyalty_member_id	object	100.0%	101,543	SKP116656
flight_date_key	object	100.0%	672	20250513
month_key	object	100.0%	24	202505
route_key	object	100.0%	289	1
fare_family	object	100.0%	3	PROMO
base_fare_brl	object	91.5%	96,536	802.48
taxes_brl	object	100.0%	36,027	224.69
booking_channel	object	100.0%	5	web
is_award_redemption	object	100.0%	2	False
points_redeemed	object	8.5%	4,612	11100
skypoints_earned	object	100.0%	3,821	1605
tier_points_earned	object	100.0%	2,274	902

fact_liability_month • 24 rows

Monthly point-liability roll-forward from the transaction ledger, with the finance sub-ledger figures alongside for tie-out.

column	dtype	non-null	distinct	example
month_key	object	100.0%	24	202409
points_issued_m	object	100.0%	24	257562484
points_expired_m	object	100.0%	8	0
points_redeemed_m	object	100.0%	24	5066662
adjust	object	100.0%	1	0
points_outstanding_eop	object	100.0%	24	252495822
liability_brl_gross	object	100.0%	24	5302412.26
liability_brl_breakage_adj	object	100.0%	24	3181447.36
points_issued_m_ledger	object	100.0%	24	257562484
points_redeemed_m_ledger	object	100.0%	24	5066662
points_expired_m_ledger	object	100.0%	8	0
breakage_provision_brl	object	100.0%	24	2120964.9
reward_cash_cost_brl	object	100.0%	24	42559.96

fact_member_month • 1,897,349 rows

Member x month panel — the retention / program-health workhorse. balance, earn/burn/expiry, tier, activity, and lapse (null until the 12-month rule is evaluable).

column	dtype	non-null	distinct	example
member_key	object	100.0%	120,000	1
member_id	object	100.0%	120,000	SKP100000
month_key	object	100.0%	24	202507
ym	object	100.0%	24	2025-07
tier_key	object	100.0%	4	1
experiment_arm	object	100.0%	3	not_enrolled
points_earned_m	object	100.0%	20,457	16243
points_redeemed_m	object	100.0%	4,687	0
points_expired_m	object	100.0%	735	0
redemptions_m	object	100.0%	2	0.0
balance_pts_eom	object	100.0%	63,231	16243
has_redeemed_ever_eom	object	100.0%	2	False
flights_m	object	100.0%	23	0
flight_revenue_brl_m	object	100.0%	147,521	0.0
card_spend_brl_m	object	100.0%	281,655	1780.85
tenure_months	object	100.0%	24	0
is_new_enrollment	object	100.0%	2	True
months_since_last_activity	object	100.0%	25	0
is_active_eom	object	100.0%	2	True
is_lapsed_eom	object	35.2%	2	0.0
became_lapsed_this_month	object	100.0%	2	False
is_reactivation	object	100.0%	2	False

fact_point_transaction • 1,593,188 rows

One SkyPoints ledger entry (earn / redeem / expire / adjust). Signed points; points_value_brl in BRL; earn_month_key on earn rows.

column	dtype	non-null	distinct	example
member_key	object	100.0%	111,624	86972
member_id	object	100.0%	111,624	SKP186971
txn_date_key	object	100.0%	672	20241003
month_key	object	100.0%	24	202410
txn_type	object	100.0%	3	earn
earn_source_key	object	94.7%	8	5.0
reward_type_key	object	3.6%	4	3.0
points	object	100.0%	15,709	127
points_value_brl	object	100.0%	14,653	2.67
earn_month_key	object	94.7%	24	202410
txn_id	object	100.0%	1,593,188	PTX001340948

fact_targets_month • 120 rows

Plan / target values per month per KPI (doc 01 section 12).

column	dtype	non-null	distinct	example
month_key	object	100.0%	24	202409
kpi	object	100.0%	5	active_members
target_value	object	100.0%	120	45000.0

fact_tier_change • 46,476 rows

Tier-change events (the movement source of truth).

column	dtype	non-null	distinct	example
member_key	object	100.0%	42,117	159
member_id	object	100.0%	42,117	SKP100158
change_date_key	object	100.0%	120	20240901
change_month_key	object	100.0%	24	202409
from_tier	object	100.0%	4	Blue
to_tier	object	100.0%	4	Gold
from_tier_key	object	100.0%	4	1
to_tier_key	object	100.0%	4	3
direction	object	100.0%	2	upgrade
trigger	object	100.0%	2	qualification

AeroVanti SkyPoints — ETL Pipeline

Document status: Phase 4 deliverable. **Depends on:** `02_data_architecture.md` . Stage-by-stage narrative of what `etl/` actually does, and the techniques used.

Run it all with:

```
python -m pip install -r etl/requirements.txt
python etl/run_pipeline.py          # 01 raw -> 02 stage -> 03 curate -> dq_checks
python etl/gen_data_dictionary.py  # regenerate docs/03_data_dictionary.md
```

Dev knobs: `SKP_SCALE` (fraction of the 120k member population, for fast iterations) and `SKP_NO_CSV=1` (skip the curated CSV mirror). Both default to the full build.

0. `config.py` — the single source of truth

One `SEED` (`20260902`). Every generator draws a **named sub-stream** from it (`utils.rng("members")` , `rng("point_engine")` , ...) so adding a draw in one place does not shift draws everywhere else. `config.py` also holds the window dates, program scale, tier / points / earn / redemption parameters, the seasonality curves, the experiment design, the DQ-injection knobs — and `FLASH_EFFECT` , the **true experiment effect**, which is applied only in `01_generate_raw.py` and never read downstream.

1. `01_generate_raw.py` — synthetic world → messy source exports

Builds the whole program in memory, vectorised over the 120k members × 24 months, then writes **source-shaped, deliberately messy** files under `data/raw/<system>/` .

1.1 Members (`build_members`)

Enrolment month from a weighted curve (relaunch surge in months 1–3, Black-Friday bumps), channel, home region/city, a latent engagement $\sim N(0,1)$, and a **status tier drawn to hit the 68/20/9/3 mix** (tilted by engagement). Flight propensity is heavy-tailed ($\Gamma \times e^{\{0.3 \cdot \text{eng}\}}$) and **boosted by tier** (Blue 1× ... Platinum 7×) so revenue per member rises steeply by tier. Members enrolled in months 1–3 are "migrated" and carry an **opening SkyPoints balance** (mixture: a big block in the 3.5k–15k "can't do much yet" band) with a lot-expiry month in the Feb–Aug 2026 wave.

1.2 Activity, tiers, earn (`build_activity`)

Monthly flight counts $\sim \text{Poisson}(\text{propensity}/12 \cdot \text{seasonality} \cdot \text{tenure-ramp})$, expanded to individual segments with a fare draw and a flexible-fare flag. **Effective tier per member-month** = Blue until `enrol + ramp_delay` , then the status tier, with ~11% of Silver+ members losing a tier and ~7% of lower members gaining one mid-window (this is what populates the tier-change ledger). SkyPoints earned per segment = `base_fare × tier multiplier × (1 + flex bonus)` . Co-brand card adoption follows a per-channel logistic; card spend is lognormal with a Nov–Dec bump; card SkyPoints = `spend / 2.5` plus a welcome bonus. Partner / promo / service-recovery earn are sparse small draws.

1.3 Point-lot engine (`run_point_engine`)

A **vectorised monthly loop** over a (`members × 25`) lot matrix (column 0 = the legacy lot, columns 1–24 = each window month's issuance). Each month:

1. add the month's earnings to the current lot column;
2. compute each member's redemption probability — $\text{logit} = A0 + 0.62 \cdot \text{eng} + 0.55 \cdot \text{tenure} + 0.40 \cdot (\text{season}-1) + 0.30 \cdot (\text{tier}-1) + 0.18 \cdot \log_{10}(\text{balance}/10k)$, gated on `balance ≥ 10,000` (the pre-Flash minimum redemption threshold);
3. for redeemers, pick a pre-Flash reward (award short / medium / points-plus-cash), `cost = price × jitter` capped at balance, and **subtract FIFO** across lot columns;
4. expire the legacy lot in its assigned month.

`A0` was calibrated so the pre-experiment steady state sits mid-band for the doc-01 KPIs.

1.4 Lapse (`build_lapse`)

A running "months since last activity" per member. `is_lapsed` = 12 consecutive months with no earn and no redemption; **left null** for the first 11 months / any member with < 12 months tenure. Reactivation = activity after a lapsed month.

1.5 The Flash Redemption experiment (`build_experiment`)

- **Eligible pool** frozen at 2026-03-01: active (earn or flight in trailing 12m) and not lapsed.
- **Stratified sample** with Gold/Platinum oversampling — {Blue 7000, Silver 3800, Gold 2600, Platinum 1600} — 50/50 within each stratum.
- **Pre-period covariates** (trailing-12m earn / flights / redemptions, balance, prior- redeemer flag, tenure) frozen at the same date.
- **Treatment incremental redemptions**: for each stratum a **deterministic count** `round(effect_pp × n_treatment)` of eligible non-redeemers (balance ≥ 3,500) is converted to redeemers, so the per-stratum incremental rate equals the baked effect up to which-members noise; the analysis still faces real binomial sampling variance. Reward mix ≈ 46% low-cost-catalog-only. Redemption **week** is drawn from a novelty-weighted distribution (week-1 hazard × 1.9, decaying with $\tau = 2.5$ weeks).
- **Guardrail outcomes** per subject: revenue (window + 8 weeks) scaled by `1 + FLASH_EFFECT.revenue_per_member_rel`; liability drawdown and partner reward cash cost from the redemptions; a 90-day post-window disengagement proxy with the baked favourable shift.
- Emits `flesk__ab_assignments`, `flesk__exposure_log`, and the reconstructable `redemption_week_feed / outcome_feed`.

1.6 Messy raw emission (`emit_raw`)

The clean internal frames are degraded into source-faithful exports — see `08_data_quality.md` for the full injection list. Highlights: SkyCore member snapshots as monthly files with ~15% duplicated rows; `point_transactions` as monthly files with a 3-day overlap into the next file; gzip files correctly named `.csv.gz`; BancoAV in Latin-1 with "R\$ 1.234,56" money strings and Excel-serial months and a re-delivered June-2025 file; Reserva in `MM/DD/YYYY` with orphan / non-member ids; Aurora with mojibake and invalid birth years and no rows for the first 3 cohorts; Flesk timestamps in UTC with 30 ineligible late corrections and at-least-once exposure rows; the sub-ledger `.xlsx` with parenthesis negatives and two restated months.

2. `02_clean_stage.py` — typed `stg_*` tables + DQ fragments

One `stg_<source>_<entity>.parquet` per raw entity. No cross-source joins, no metrics. Vectorised parsers:

Parser	Handles
<code>parse_dates(hint)</code>	ISO, <code>DD/MM/YYYY</code> , <code>MM/DD/YYYY</code> (per-source locale rule), Excel serials (epoch 1899-12-30).
<code>parse_money_br1</code>	R\$, . -thousands / , -decimal, (x) → -x, blank → null (never 0). <code>pd.NA</code> -safe.
<code>parse_points</code>	text magnitudes ("1.500"), sign by <code>txn_type</code> .
<code>map_vocab</code>	controlled vocabularies in <code>et1/vocab/*.csv</code> ; unknowns → explicit "unknown" + counted.
<code>fix_encoding</code>	<code>ftfy.fix_text</code> + BOM strip for BancoAV / Aurora.

De-duplication: SkyCore member snapshots collapsed to one row per member per month; `point_transactions` deduped on `txn_id` across the file overlap; the re-delivered BancoAV month dropped on (`card_id`, `competencia_month`); at-least-once event/exposure rows dropped; the ledger's restated months resolved to the last occurrence. Every fix increments a counter that lands in `data/quality/dq_fragments/<entity>.json`.

3. `03_build_curated.py` — the Kimball star + experiment tables

Reads only staging. `dim_date` first (2024-01-01 → 2027-12-31; carries `experiment_phase`, `experiment_week`, BR holidays, season tags).

- **dim_member** — latest snapshot per member + card / CRM enrichment + `is_steady_state_cohort`
- experiment arm/stratum + **ledger-derived current tier**.
- **fact_tier_change** — deduped tier-change events (incl. the initial Blue → status qualification event); **curated tier status is folded forward from this ledger**, and the monthly tier snapshot is used only to compute a drift metric.
- **fact_point_transaction** — every ledger entry with signed points, `points_value_br1`, `earn_month_key` on earn rows, and resolved dimension keys (-2 for orphan members).

- `fact_flight_segment` — one flown segment; `member_key` -1 non-member / -2 orphan; `skypoints_earned` recomputed with the month's ledger tier multiplier; a slim `dim_route`.
- `fact_card_spend_month`, `fact_liability_month` (transaction roll-forward with the finance sub-ledger alongside for the tie-out), `fact_targets_month`.
- `fact_member_month` — the panel: balance (cumulative earn – redeem – expire), earn/burn/expiry, tier, flights & revenue, card spend, `has_redeemed_ever`, `months_since_last_activity` (vectorised forward-fill), `is_active_eom`, `is_lapsed_eom` (null until evaluable), `became_lapsed_this_month`, `is_reactivation`.
- `fact_experiment_assignment` / `_member_week` / `_outcome` — reconciled to eligible subjects; the outcome table pre-computes the primary metric, the four guardrails and the diagnostics so Phase 10 goes straight to the tests.

Cross-cutting: FIFO lot accounting in the engine; breakage-adjusted liability unit cost; sentinel keys never dropped; all money BRL.

4. `dq_checks.py` — hard assertions + `dq_report.md`

Fails the build (non-zero exit) on any hard violation. Checks: key uniqueness and referential integrity; **points roll-forward identity** (program balance == Σ member balance, exact); **liability tie-out** to the sub-ledger; lapse null-honesty and stickiness; experiment integrity (assignment count, subject membership, 50/50 allocation, `redeemed_in_window` == OR of weekly flags, $\geq$ 90-day post horizon); and the doc-01 benchmark bands on the pre-experiment steady state (redemption rate, ever-redeemed share, burn/earn ratio, Blue share). Then it assembles every DQ fragment into `data/quality/dq_report.md`.

AeroVanti SkyPoints — Data Model

Document status: Phase 4 deliverable. **Depends on:** 02_data_architecture.md , 03_data_dictionary.md (auto-generated).
Describes the curated star as it will be imported into the Power BI semantic model (Phase 6).

1. Bus matrix (facts × dimensions)

Fact ↓ / Dim →	date	member	tier	earn_source	reward_type	route	experiment_arm	experiment_stratum
fact_point_transaction	•	•	(via member)	•	•		(via member)	
fact_member_month	• (month)	•	•				•	(via member)
fact_flight_segment	•	•	•			•	(via member)	
fact_card_spend_month	• (month)	•						
fact_tier_change	•	•	• (from/to)					
fact_liability_month	• (month)							
fact_experiment_assignment	• (assigned_ts)	•					•	•
fact_experiment_member_week	• (exp. week)	•					•	•
fact_experiment_outcome		•					•	•
fact_targets_month	• (month)							

• = direct relationship; (via ...) = reachable through another dimension, not a physical relationship.

2. Fact tables — grain & measures

Fact	Grain	Additive measures	Non-additive / semi-additive
fact_point_transaction	one ledger entry	points (signed), points_value_brl	—
fact_member_month	member × calendar month	points_earned_m , points_redeemed_m , points_expired_m , redemptions_m , flights_m , flight_revenue_brl_m , card_spend_brl_m	balance_pts_eom (semi-additive, last), is_active_eom , is_lapsed_eom (nullable), has_redeemed_ever_eom , tier_key , months_since_last_activity
fact_flight_segment	one flown segment	base_fare_brl , taxes_brl , skypoints_earned , tier_points_earned , points_redeemed	is_award_redemption
fact_card_spend_month	member × month	card_spend_brl , skypoints_earned	is_card_active
fact_tier_change	one tier-change event	count of events	direction , trigger , from_tier_key , to_tier_key
fact_liability_month	calendar month	points_issued_m , points_redeemed_m , points_expired_m , breakage_provision_brl , reward_cash_cost_brl	points_outstanding_eop , liability_brl_gross , liability_brl_breakage_adj (all semi-additive, last) + *_ledger tie-out columns

Fact	Grain	Additive measures	Non-additive / semi-additive
fact_experiment_assignment	subject (member)	count of subjects	pre_earn_12m_pts , pre_flights_12m , pre_redemptions_12m , pre_balance_pts , prior_redeemer_flag , pre_tenure_months
fact_experiment_member_week	subject × experiment week (1–12)	redemptions_w , points_redeemed_w , redemption_value_brl_w	redeemed_w , low_cost_only_w
fact_experiment_outcome	subject (member)	redemptions_in_window , redemption_value_brl , points_redeemed_in_window , revenue_brl_window_plus8w , liability_drawdown_brl , reward_cash_cost_brl , net_liability_cost_brl , flights_in_window	redeemed_in_window (primary metric), low_cost_only , disengaged_90d_post , earn_in_window , active_post_quarter
fact_targets_month	month × KPI	target_value	kpi

3. Dimensions — key attributes

Dimension	Key	Attributes used in the report
dim_date	date_key (int YYYYMMDD)	year , quarter , month , month_key , month_name , month_sort , iso_week , is_month_end , is_holiday_br , season_tag , experiment_phase , experiment_week
dim_member	member_key	enrollment_cohort_month , enrollment_channel , home_region , home_city , has_cobrand_card , current_tier_name , crm_lifecycle_stage , age_band , is_steady_state_cohort , experiment_arm , experiment_stratum , is_experiment_subject
dim_tier	tier_key	tier_name , tier_rank , earn_multiplier , is_elite
dim_earn_source	earn_source_key	source_name , source_group , earns_tier_points
dim_reward_type	reward_type_key	reward_name , reward_category , points_price_typical , value_per_point_brl , is_low_cost_catalog
dim_route	route_key	origin , dest , haul_band , region_pair
dim_experiment_arm	arm_key	(control / treatment / not_enrolled) — disconnected slicer helper
dim_experiment_stratum	stratum	(Blue ... Platinum) — disconnected slicer helper

Sentinel members: member_key -1 (non-member booking), -2 (orphan). Sentinel -1 rows also exist in dim_earn_source , dim_reward_type , dim_route .

4. Relationships

From (fact)	Column	To (dim)	Column	Card.	Cross-filter	Active
fact_point_transaction	member_key	dim_member	member_key	*:1	single	yes
fact_point_transaction	txn_date_key	dim_date	date_key	*:1	single	yes
fact_point_transaction	earn_source_key	dim_earn_source	earn_source_key	*:1	single	yes
fact_point_transaction	reward_type_key	dim_reward_type	reward_type_key	*:1	single	yes
fact_member_month	member_key	dim_member	member_key	*:1	single	yes

From (fact)	Column	To (dim)	Column	Card.	Cross-filter	Active
fact_member_month	month_key	dim_month	month_key	* : 1	single	yes
fact_member_month	tier_key	dim_tier	tier_key	* : 1	single	yes
fact_flight_segment	member_key	dim_member	member_key	* : 1	single	yes
fact_flight_segment	flight_date_key	dim_date	date_key	* : 1	single	yes
fact_flight_segment	route_key	dim_route	route_key	* : 1	single	yes
fact_card_spend_month	member_key	dim_member	member_key	* : 1	single	yes
fact_card_spend_month	month_key	dim_month	month_key	* : 1	single	yes
fact_tier_change	member_key	dim_member	member_key	* : 1	single	yes
fact_tier_change	change_date_key	dim_date	date_key	* : 1	single	yes
fact_liability_month	month_key	dim_month	month_key	* : 1	single	yes
fact_experiment_assignment	member_key	dim_member	member_key	1 : 1	both	yes
fact_experiment_member_week	member_key	dim_member	member_key	* : 1	single	yes
fact_experiment_outcome	member_key	dim_member	member_key	1 : 1	single	yes
fact_targets_month	month_key	dim_month	month_key	* : 1	single	yes

dim_experiment_arm / dim_experiment_stratum are **disconnected** — used only to drive slicers whose selections are read by measures with SELECTEDVALUE . The physical arm/stratum columns live on dim_member and the experiment facts.

Date handling. dim_date is at **day** grain (its month_key is *not* unique, so it cannot sit on the "one" side of any relationship). A small dim_month helper table (one row per month, unique month_key) is the join target for the month-grain facts (fact_member_month , fact_card_spend_month , fact_liability_month , fact_targets_month); day-grain facts join dim_date[date_key] . There is deliberately **no** dim_month ↔ dim_date relationship. Time-series visuals bind to dim_month[month_name] (sorted by month_sort) for chronological order.

5. Measure groups (Phase 7 preview)

Program Health · Earn & Burn · Breakage & Liability · Tiers & Value · Retention & Lapse · Commercial Value · **Experiment** (redemption rate by arm, lift + CI + p, SRM, weekly series, guardrails, per-tier effect) · Time Intelligence (YoY, YTD, MAT, steady-state cut) · Targets vs Plan · a dynamic Exec Insight narrative measure.

AeroVanti SkyPoints — DAX Measure Library

Auto-generated from `etL/pbi_measures.py` on 2026-09-02 by `etL/gen_dax_doc.py`. Do not edit by hand.

62 measures in the `_Measures` table, injected into the semantic model by `powerbi/gen_semantic_model.py`.

Conventions

- All money is **BRL** — no currency slicer.
- "Latest month" = `MAX(dim_month[month_key])` in the current filter context.
- Time intelligence uses `dim_month[month_sort]` (a linear `year*12 + month` index): `PY = month_sort - 12`. No marked date table, no `SAMEPERIODYEAR`.
- Experiment measures read `fact_experiment_outcome / fact_experiment_member_week` and respect the Experiment Arm / Stratum slicers; arm-specific measures pin the arm with `CALCULATE(..., arm_key = "...")`.
- **Breakage rate, lapse rate and liability per member are lower-is-better** — their `vs Target` measures are good when negative.

01 Program

Members

format `#,0`

Enrolled members, excluding the two sentinel rows.

```
CALCULATE ( DISTINCTCOUNT ( dim_member[member_key] ), dim_member[member_key] > 0 )
```

Active Members

format `#,0`

Members with an earn or flight in the trailing 12 months, at the latest month in context.

```
VAR mk = MAX ( dim_month[month_key] )
RETURN
CALCULATE (
    DISTINCTCOUNT ( fact_member_month[member_key] ),
    fact_member_month[is_active_eom] = TRUE (),
    fact_member_month[month_key] = mk
)
```

Active Rate %

format `0.0%`

```
DIVIDE ( [Active Members], [Members] )
```

New Enrolments

format `#,0`

First enrolment month for a member.

```
CALCULATE ( COUNTROWS ( fact_member_month ), fact_member_month[is_new_enrollment] = TRUE () )
```

Co-brand Card Penetration %

format `0.0%`

```
DIVIDE (
    CALCULATE ( DISTINCTCOUNT ( dim_member[member_key] ),
        dim_member[has_cobrand_card] = TRUE (), dim_member[member_key] > 0 ),
    [Members]
)
```

02 Earn & Burn

SkyPoints Issued

format #,0

```
CALCULATE ( SUM ( fact_point_transaction[points] ), fact_point_transaction[txn_type] = "earn" )
```

SkyPoints Redeemed

format #,0

```
- CALCULATE ( SUM ( fact_point_transaction[points] ), fact_point_transaction[txn_type] = "redeem" )
```

SkyPoints Expired

format #,0

```
- CALCULATE ( SUM ( fact_point_transaction[points] ), fact_point_transaction[txn_type] = "expire" )
```

Burn Earn Ratio

format 0.0%

Points redeemed as a share of points issued over the period in context.

```
DIVIDE ( [SkyPoints Redeemed], [SkyPoints Issued] )
```

Redemptions

format #,0

```
CALCULATE ( COUNTROWS ( fact_point_transaction ), fact_point_transaction[txn_type] = "redeem" )
```

Redeeming Members

format #,0

```
CALCULATE ( DISTINCTCOUNT ( fact_point_transaction[member_key] ), fact_point_transaction[txn_type] = "redeem" )
```

Quarterly Redemption Rate %

format 0.0%

Share of active members with >= 1 redemption in the trailing quarter.

```
VAR mk = MAX ( dim_month[month_key] )
VAR q = FILTER ( ALL ( dim_month ),
    dim_month[month_sort] > MAXX ( FILTER ( ALL ( dim_month ), dim_month[month_key] = mk ), dim_month[month_key] )
    && dim_month[month_key] <= mk )
VAR redeemers =
    CALCULATE ( DISTINCTCOUNT ( fact_member_month[member_key] ),
```

```

fact_member_month[redemptions_m] > 0, q )
VAR active =
    CALCULATE ( DISTINCTCOUNT ( fact_member_month[member_key] ),
        fact_member_month[is_active_eom] = TRUE (), fact_member_month[month_key] = mk )
RETURN DIVIDE ( redeemers, active )

```

Ever-Redeemed Share %

format 0.0%

```

VAR mk = MAX ( dim_month[month_key] )
RETURN
DIVIDE (
    CALCULATE ( DISTINCTCOUNT ( fact_member_month[member_key] ),
        fact_member_month[has_redeemed_ever_eom] = TRUE (),
        fact_member_month[month_key] = mk ),
    [Active Members]
)

```

Avg Points Balance

format #,0

```

VAR mk = MAX ( dim_month[month_key] )
RETURN
CALCULATE ( AVERAGE ( fact_member_month[balance_pts_eom] ),
    fact_member_month[month_key] = mk, fact_member_month[is_active_eom] = TRUE () )

```

03 Breakage & Liability

Point Liability BRL

format "R\$"#,0

```

VAR mk = MAX ( dim_month[month_key] )
RETURN CALCULATE ( SUM ( fact_liability_month[liability_brl_gross] ),
    fact_liability_month[month_key] = mk )

```

Point Liability BRL (breakage-adj)

format "R\$"#,0

```

VAR mk = MAX ( dim_month[month_key] )
RETURN CALCULATE ( SUM ( fact_liability_month[liability_brl_breakage_adj] ),
    fact_liability_month[month_key] = mk )

```

Breakage Rate %

format 0.0%

Points expired as a share of points issued over the period in context (realised).

```

DIVIDE ( [SkyPoints Expired], [SkyPoints Issued] )

```

Points Expiring BRL

format "R\$"#,0

```
SUM ( fact_liability_month[points_expired_m] ) * 0.021
```

Liability per Active Member

format "R\$"#,0.0

```
DIVIDE ( [Point Liability BRL], [Active Members] )
```

04 Tiers / 05 Commercial

Members in Tier

format #,0

```
VAR mk = MAX ( dim_month[month_key] )
RETURN CALCULATE ( DISTINCTCOUNT ( fact_member_month[member_key] ),
    fact_member_month[month_key] = mk, fact_member_month[is_active_eom] = TRUE ( ) )
```

Elite Share %

format 0.0%

```
DIVIDE (
    CALCULATE ( [Members in Tier], dim_tier[is_elite] = TRUE ( ) ),
    CALCULATE ( [Members in Tier], REMOVEFILTERS ( dim_tier ) )
)
```

Tier Upgrades

format #,0

```
CALCULATE ( COUNTROWS ( fact_tier_change ), fact_tier_change[direction] = "upgrade" )
```

Tier Downgrades

format #,0

```
CALCULATE ( COUNTROWS ( fact_tier_change ), fact_tier_change[direction] IN { "downgrade", "retain_drop" } )
```

Flight Revenue BRL

format "R\$"#,0

```
CALCULATE ( SUM ( fact_flight_segment[base_fare_br1] ), fact_flight_segment[is_award_redemption] = FALSE ( ) )
```

Revenue per Active Member

format "R\$"#,0

```
DIVIDE ( [Flight Revenue BRL], [Active Members] )
```

Award Seat Share %

format 0.0%

```
DIVIDE (
    CALCULATE ( COUNTROWS ( fact_flight_segment ), fact_flight_segment[is_award_redemption] = TRUE ( ) ),
    CALCULATE ( COUNTROWS ( fact_flight_segment ) )
)
```

```
COUNTROWS ( fact_flight_segment )
)
```

Card Spend BRL

format "R\$"#,0

```
SUM ( fact_card_spend_month[card_spend_brl] )
```

06 Retention & Lapse

Account-Lapsed Members

format #,0

Members in account-lapse (no earn and no redemption for 12 months) at the latest month.

```
VAR mk = MAX ( dim_month[month_key] )
RETURN CALCULATE ( DISTINCTCOUNT ( fact_member_month[member_key] ),
                    fact_member_month[is_lapsed_eom] = 1, fact_member_month[month_key] = mk )
```

Account Lapse Rate %

format 0.0%

```
DIVIDE ( [Account-Lapsed Members], [Active Members] + [Account-Lapsed Members] )
```

Engagement-Lapsed Members

format #,0

Members with no flight and no redemption for 12 months (card-only earn does not count).

```
VAR mk = MAX ( dim_month[month_key] )
VAR startsort = MAXX ( FILTER ( ALL ( dim_month ), dim_month[month_key] = mk ), dim_month[month_sort] ) - 11
RETURN
COUNTROWS (
    FILTER (
        VALUES ( fact_member_month[member_key] ),
        CALCULATE (
            SUM ( fact_member_month[flights_m] ) + SUM ( fact_member_month[redemptions_m] ),
            FILTER ( ALL ( dim_month ),
                    dim_month[month_sort] >= startsort
                    && dim_month[month_key] <= mk )
        ) = 0
    )
)
```

Engagement Lapse Rate %

format 0.0%

```
DIVIDE ( [Engagement-Lapsed Members], [Active Members] )
```

Reactivations

format #,0

```
CALCULATE ( COUNTROWS ( fact_member_month ), fact_member_month[is_reactivation] = TRUE () )
```

07 A/B Test

Exp Subjects

format #,0

```
DISTINCTCOUNT ( fact_experiment_outcome[member_key] )
```

Exp Redemption Rate

format 0.0%

Share of subjects (in the current arm/stratum filter) with >= 1 redemption in the 12-week window.

```
DIVIDE (
    CALCULATE ( COUNTROWS ( fact_experiment_outcome ),
        fact_experiment_outcome[redeemed_in_window] = TRUE () ),
    COUNTROWS ( fact_experiment_outcome )
)
```

Exp Control Rate

format 0.0%

```
CALCULATE ( [Exp Redemption Rate], fact_experiment_outcome[arm_key] = "control" )
```

Exp Treatment Rate

format 0.0%

```
CALCULATE ( [Exp Redemption Rate], fact_experiment_outcome[arm_key] = "treatment" )
```

Exp Lift pp

format +0.00;-0.00;0.00

Absolute treatment - control difference in the redemption rate, in percentage points.

```
( [Exp Treatment Rate] - [Exp Control Rate] ) * 100
```

Exp Lift Rel %

format +0.0%;-0.0%;0.0%

```
DIVIDE ( [Exp Treatment Rate] - [Exp Control Rate], [Exp Control Rate] )
```

Exp Lift CI Low pp

format +0.00;-0.00;0.00

```
VAR p1 = [Exp Control Rate]  VAR p2 = [Exp Treatment Rate]
VAR n1 = CALCULATE ( [Exp Subjects], fact_experiment_outcome[arm_key] = "control" )
VAR n2 = CALCULATE ( [Exp Subjects], fact_experiment_outcome[arm_key] = "treatment" )
VAR se = SQRT ( DIVIDE ( p1 * ( 1 - p1 ), n1 ) + DIVIDE ( p2 * ( 1 - p2 ), n2 ) )
RETURN ( ( p2 - p1 ) - 1.959964 * se ) * 100
```

Exp Lift CI High pp

format +0.00;-0.00;0.00

```
VAR p1 = [Exp Control Rate]  VAR p2 = [Exp Treatment Rate]
VAR n1 = CALCULATE ( [Exp Subjects], fact_experiment_outcome[arm_key] = "control" )
VAR n2 = CALCULATE ( [Exp Subjects], fact_experiment_outcome[arm_key] = "treatment" )
VAR se = SQRT ( DIVIDE ( p1 * ( 1 - p1 ), n1 ) + DIVIDE ( p2 * ( 1 - p2 ), n2 ) )
RETURN ( ( p2 - p1 ) + 1.959964 * se ) * 100
```

Exp Z Stat

format 0.00

Two-proportion pooled z-statistic.

```
VAR p1 = [Exp Control Rate]  VAR p2 = [Exp Treatment Rate]
VAR n1 = CALCULATE ( [Exp Subjects], fact_experiment_outcome[arm_key] = "control" )
VAR n2 = CALCULATE ( [Exp Subjects], fact_experiment_outcome[arm_key] = "treatment" )
VAR pp = DIVIDE ( p1 * n1 + p2 * n2, n1 + n2 )
VAR se = SQRT ( pp * ( 1 - pp ) * ( DIVIDE ( 1, n1 ) + DIVIDE ( 1, n2 ) ) )
RETURN DIVIDE ( p2 - p1, se )
```

Exp Significant 95%

```
IF ( ABS ( [Exp Z Stat] ) > 1.959964, "Significant", "Not significant" )
```

Exp SRM Ratio

format 0.0%

Treatment share of subjects; expect 50% (sample-ratio-mismatch check).

```
DIVIDE (
    CALCULATE ( [Exp Subjects], fact_experiment_outcome[arm_key] = "treatment" ),
    [Exp Subjects]
)
```

Exp Revenue per Member

format "R\$"#,0

```
AVERAGE ( fact_experiment_outcome[revenue_brl_window_plus8w] )
```

Exp Revenue per Member Delta %

format +0.0%;-0.0%;0.0%

Guardrail: non-inferiority margin is -3%.

```
DIVIDE (
    CALCULATE ( [Exp Revenue per Member], fact_experiment_outcome[arm_key] = "treatment" )
    - CALCULATE ( [Exp Revenue per Member], fact_experiment_outcome[arm_key] = "control" ),
    CALCULATE ( [Exp Revenue per Member], fact_experiment_outcome[arm_key] = "control" )
)
```

Exp Net Liability Cost per Member

format "R\$"#,0.00

```
AVERAGE ( fact_experiment_outcome[net_liability_cost_br1] )
```

Exp Disengaged 90d %

format 0.0%

Guardrail: share with no earn and no redemption in the 90 days after the window.

```
DIVIDE (
  CALCULATE ( COUNTROWS ( fact_experiment_outcome ),
    fact_experiment_outcome[disengaged_90d_post] = TRUE ( ) ),
  COUNTROWS ( fact_experiment_outcome )
)
```

Exp Redemption Value per Redeemer

format "R\$"#,0

Guardrail: depth of redemptions (Flash makes them shallower).

```
DIVIDE (
  SUM ( fact_experiment_outcome[redemption_value_br1] ),
  CALCULATE ( COUNTROWS ( fact_experiment_outcome ),
    fact_experiment_outcome[redeemed_in_window] = TRUE ( ) )
)
```

Exp Weekly Redeemers %

format 0.00%

Redeemers in the week / subjects — plot by week and arm for the novelty check.

```
DIVIDE ( CALCULATE ( COUNTROWS ( fact_experiment_member_week ), fact_experiment_member_week[redeemed_w] = TRUE ( )
```

Exp Effect by Tier pp

format +0.00;-0.00;0.00

Same as Exp Lift pp; place on a bar by Experiment Stratum for the heterogeneity cut.

[Exp Lift pp]

08 Time & Targets

SkyPoints Redeemed PY

format #,0

```
CALCULATE (
  [SkyPoints Redeemed],
  FILTER ( ALL ( dim_month ),
    dim_month[month_sort] = MAX ( dim_month[month_sort] ) - 12 )
)
```

SkyPoints Redeemed YoY %

format +0.0%;-0.0%;0.0%

```
DIVIDE ( [SkyPoints Redeemed] - [SkyPoints Redeemed PY], [SkyPoints Redeemed PY] )
```

Breakage Rate Target

format 0.0%

```
VAR mk = MAX ( dim_month[month_key] )
RETURN CALCULATE ( AVERAGE ( fact_targets_month[target_value] ),
                    fact_targets_month[kpi] = "breakage_rate", fact_targets_month[month_key] = mk )
```

Redemption Rate Target

format 0.0%

```
VAR mk = MAX ( dim_month[month_key] )
RETURN CALCULATE ( AVERAGE ( fact_targets_month[target_value] ),
                    fact_targets_month[kpi] = "redemption_rate_quarterly", fact_targets_month[month_key] = mk )
```

Quarterly Redemption Rate vs Target

format +0.0%;-0.0%;0.0%

```
[Quarterly Redemption Rate %] - [Redemption Rate Target]
```

Breakage Rate vs Target

format +0.0%;-0.0%;0.0%

Lower is better — negative is good.

```
[Breakage Rate %] - [Breakage Rate Target]
```

09 Narrative

Exec Insight

Dynamic summary sentence for the Executive Summary page; updates with the slicers.

```
VAR lift = [Exp Lift pp]
VAR sig = [Exp Significant 95%]
VAR br = [Breakage Rate %]
VAR rr = [Quarterly Redemption Rate %]
RETURN
"Flash Redemption lifted the 12-week redemption rate by "
& FORMAT ( lift, "+0.0;-0.0" ) & " pp ( " & sig & " at 95% control "
& FORMAT ( [Exp Control Rate], "0.0%" ) & " vs treatment "
& FORMAT ( [Exp Treatment Rate], "0.0%" ) & "). "
& "Program-wide, the quarterly redemption rate is " & FORMAT ( rr, "0.0%" )
& " and realised breakage is " & FORMAT ( br, "0.0%" )
& " of points issued, against a ~40% lifetime expectation. "
& "Guardrails: revenue per member " & FORMAT ( [Exp Revenue per Member Delta %], "+0.0%;-0.0%" )
& ", 90-day disengagement "
& FORMAT ( CALCULATE ( [Exp Disengaged 90d %], fact_experiment_outcome[arm_key] = "treatment" )
- CALCULATE ( [Exp Disengaged 90d %], fact_experiment_outcome[arm_key] = "control" ), "+0.0%;-0.0%" )
& "."
```

10 Funnel

Members with Redeemable Balance

format #,0

Active members whose balance clears the Flash minimum redemption threshold.

```
VAR mk = MAX ( dim_month[month_key] )
RETURN CALCULATE ( DISTINCTCOUNT ( fact_member_month[member_key] ),
    fact_member_month[month_key] = mk,
    fact_member_month[is_active_eom] = TRUE (),
    fact_member_month[balance_pts_eom] >= 3500 )
```

Redeemers in Last 12m

format #,0

```
VAR mk = MAX ( dim_month[month_key] )
VAR startsort = MAXX ( FILTER ( ALL ( dim_month ), dim_month[month_key] = mk ), dim_month[month_sort] ) - 11
RETURN
CALCULATE ( DISTINCTCOUNT ( fact_member_month[member_key] ),
    fact_member_month[redemptions_m] > 0,
    FILTER ( ALL ( dim_month ), dim_month[month_sort] >= startsort && dim_month[month_key] <= mk ) )
```

Redeemers in Last Quarter

format #,0

```
VAR mk = MAX ( dim_month[month_key] )
VAR startsort = MAXX ( FILTER ( ALL ( dim_month ), dim_month[month_key] = mk ), dim_month[month_sort] ) - 2
RETURN
CALCULATE ( DISTINCTCOUNT ( fact_member_month[member_key] ),
    fact_member_month[redemptions_m] > 0,
    FILTER ( ALL ( dim_month ), dim_month[month_sort] >= startsort && dim_month[month_key] <= mk ) )
```

Ever-Redeemed Members

format #,0

```
VAR mk = MAX ( dim_month[month_key] )
RETURN CALCULATE ( DISTINCTCOUNT ( fact_member_month[member_key] ),
    fact_member_month[has_redeemed_ever_eom] = TRUE (), fact_member_month[month_key] = mk )
```

Funnel Value

format #,0

Member count at the selected redemption-funnel stage (place 'Funnel Stage'[stage] on Category).

```
VAR s = SELECTEDVALUE ( 'Funnel Stage'[stage] )
RETURN
SWITCH ( s,
    "Active members", [Active Members],
    "Have points >= Flash threshold", [Members with Redeemable Balance],
    "Ever redeemed", [Ever-Redeemed Members],
    "Redeemed in last 12m", [Redeemers in Last 12m],
    "Redeemed in last quarter", [Redeemers in Last Quarter],
    BLANK () )
```

AeroVanti SkyPoints — Visual Identity

Document status: Phase 5 deliverable. The single source of truth is `assets/brand.py` , imported by `assets/gen_logo.py` and `powerbi/gen_report_visuals.py` so the logo, page backgrounds, theme and every scripted visual override draw from the same values.

Tone

Corporate, restrained, CFO-facing. Aviation-adjacent, not a marketing page — this is an A/B test read-out and a program-health dashboard.

Palette

Role	Hex	Use
AeroVanti blue	#0B5FA5	brand, primary series, table accent
Navy	#0A2A43	header band, darkest ink
Amber (accent)	#E08A1E	treatment arm · current period · "look here"
Steel (comparison)	#5C8CB8	control arm · prior year · target
Page ground	#EEF3F8	light body
Card	FFFFFF	
Gridline / hairline	#E4E9EF / #D3DBE3	quiet
Good / Bad / Neutral	#2E7D5B / #C0453B / #8A8A99	semantic; breakage / lapse / liability are lower-is-better

Categorical ramp: blue, steel, amber, teal #3E9C93 , plum #8E5EA8 , grey #6B7280 , rose #C77CA6 , indigo #4C6EA6 . Tier ramp: Blue #4C6EA6 , Silver #9AA6B2 , Gold #C9922E , Platinum #5B6B7A .

Type

Segoe UI family. Semibold for titles / callouts, Regular for body / labels. Callout (KPI value) 30 px, title 12–15 px, label 9 px.

Signature element (every page)

Navy header band (58 px) with the **AeroVanti** wordmark (amber "Vanti"), an amber 3 px hairline under it, a right-aligned "SkyPoints — Loyalty Analytics" label, then the page title + one-line question on the light body. Below: a slicer row (Analysis period · Experiment arm · Tier stratum), a white rounded KPI frame with 6 value cards, then two content rows on the 3-30-300 gradient.

Assets emitted (`assets/gen_logo.py` → `powerbi/assets/`)

- `logo_mark_on_light.png` / `_on_dark.png` — rounded-square tile, upward chevron ("sky") + an amber points dot.
- `wordmark_on_light.png` / `_on_dark.png` — "AeroVanti" lockup.
- `bg_01_executive.png` (1280×860) · `bg_02_tiers.png` · `bg_03_experiment.png` · `bg_04_funnel.png` (1280×720) — full-page backgrounds carrying the band + title.
- `_page_heights.json` .

Theme

`powerbi/theme/theme.json` — `dataColors` , semantic colours, 4 text classes, and `visualStyles` for ~20 visual types + `page` : white card background, 8 px rounded border, no drop shadow on cards / KPIs / frames, a soft shadow on charts via the `*` wildcard, data labels on for bar / column / line / combo / waterfall, gridlines only on the value axis. `pbir theme validate` passes.

AeroVanti SkyPoints — Data Quality

Document status: Phase 4 deliverable. Where each injected issue is handled, what the generated `data/quality/dq_report.md` contains, and the accepted limitations.

The pipeline **injects** realistic source defects in `01_generate_raw.py`, **handles** them in `02_clean_stage.py` / `03_build_curated.py` with a counter for every fix, and **asserts** the invariants in `dq_checks.py` (non-zero exit on any hard failure).

1. The six primary issue classes

#	Issue	Injected in	Handled in	Hard assertion
Q01	Multiple date formats + Excel serials	SkyCore <code>member_since</code> , BancoAV, Reserva (<code>MM/DD/YYYY</code>), Ledger, reward catalog	<code>parse_dates(hint)</code> — per-source locale rule + Excel-serial epoch 1899-12-30	<code>dates_unparseable = 0</code> on the key columns
Q02	Money as locale strings / parenthesis negatives / symbol-in- value	BancoAV <code>card_spend</code> , Ledger <code>.xlsx</code> , some SkyCore <code>value_br1</code>	<code>parse_money_br1</code> — strips <code>R\$</code> , <code>.</code> -thousands / , -decimal, <code>(x)</code> → <code>-x</code> , blank→null	no non-null money column is object dtype; <code>liability_br1_gross ≥ 0</code>
Q03	Encoding damage (Latin-1 mojibake, BOM)	BancoAV names/city, Aurora <code>cidade</code>	<code>ftfy.fix_text</code> + BOM strip	no <code>Ã / Â</code> mojibake sequences remain in <code>dim_member.home_city</code>
Q04	Duplicate rows (daily full-dump snapshot, overlapping monthly <code>point_transactions</code> , re- delivered BancoAV month, at-least- once event/exposure rows)	SkyCore, BancoAV, Flesk	collapse to one row per member per month · dedupe on <code>txn_id</code> · drop re- delivered file on (<code>card_id</code> , <code>month</code>) · dedupe events/exposures	<code>fact_point_transaction.txn_id</code> unique · <code>fact_flight_segment.segment_id</code> unique
Q05	Missing master rows / orphan keys (bookings → members absent from SkyCore, point rows for unknown members, months-1–3 cohorts with no CRM row)	Reserva, SkyCore, Aurora late go-live	sentinel keys (<code>-1</code> non- member, <code>-2</code> orphan) + count; CRM attributes left null with <code>crm_present</code> flag	every fact FK resolves to a real or sentinel key
Q06	Snapshot-vs-ledger drift (<code>tier_snapshots_monthly</code> disagrees with <code>tier_change_events</code>)	SkyCore	curated tier status folded forward from the event ledger (incl. an initial Blue→status qualification event); snapshot used only to measure drift	curated tier status == ledger-derived by construction ; drift rate reported (target 3– 6%)

Also handled, without ceremony

Null / placeholder keys → sentinels · invalid birth years (`1900-01-01`, `0`) → null + `birth_year_valid = false` · Flesk UTC timestamps → tz-aware, Brazil-local calendar day for `*_date_key` · ledger restated months → last occurrence wins · free-text `txn_type` / lifecycle stage / booking channel / tier labels → controlled vocab in `et1/vocab/`.

2. Accounting identities asserted (`dq_checks.py`)

- **Points roll-forward** (program-wide, per month): opening + earned – redeemed – expired = closing, exact. Program balance == $\sum$ member balance.
- **Liability tie-out:** the transaction-ledger roll-up matches the finance sub-ledger's issued / redeemed / expired within tolerance.

- **Lapse honesty:** `is_lapsed_eom` is null (never `False`) for tenure < 12 months; a lapsed member stays lapsed until a reactivation row.
 - **Experiment integrity:** exactly 15,000 assigned subjects; every weekly / outcome row is an assigned subject; 50/50 allocation within every stratum; `redeemed_in_window` == OR of the 12 weekly flags; data extends ≥ 90 days past the experiment end.
 - **Benchmark bands** on the pre-experiment steady-state cohort: quarterly redemption rate, ever-redeemed share, burn/earn ratio and the Blue share of active members all land inside their doc-01 §11 ranges.
-

3. Generated report

`data/quality/dq_report.md` (built by `dq_checks.py`) contains:

1. the pass/fail list for every assertion above;
2. a per-entity counter block from the `data/quality/dq_fragments/*.json` files — e.g. how many Excel serials were converted, mojibake strings repaired, duplicate transactions dropped, snapshot rows collapsed, orphan keys sentinelled, restated months resolved.

(latest run summary is pasted in §5 once the final pipeline lands.)

4. Accepted limitations

- **No per-lot FIFO attribution in curated.** The point-lot engine consumes lots FIFO, but `fact_point_transaction` only carries `earn_month_key` on `earn` rows. Cohort breakage in Phase 10 is computed as an aggregate roll-forward (issued vs redeemed vs expired by calendar month) plus a legacy-vs-windowed split, not a full per-lot ledger. This was a deliberate simplification under the Phase-0 "reduced ETL" decision.
- **90-day lapse proxy only.** The experiment sits in months 19–21 and the data ends at month 24, so the guardrail uses a 90-day post-window disengagement proxy, not a true 12-month lapse outcome.
- **Synthetic data.** All members, flights, card spend, earn and redemption events are seed-generated. Benchmark ranges are directional, from public airline-loyalty disclosures, and are used only to set targets.
- **Contamination not modelled.** Household / referral spillover between experiment arms is assumed negligible.

AeroVanti SkyPoints — Report Structure

Document status: Phase 8 deliverable. The Power BI report is the **secondary** deliverable (Phase-0 decision) — 4 pages, lean formatting. One command rebuilds it:

```
cd powerbi && bash build_report.sh      # SKP_ALLOW_DESKTOP=1 if Desktop has another report open
```

Pipeline: `gen_logo.py` → `gen_semantic_model.py` (+ `pbi_measures.py`) → `gen_dax_doc.py` → `gen_report_visuals.py` (emits `_report_layout.json`) → `_drive_pbir.py` (drives `pbir`). Output: `powerbi/AeroVantiSkyPoints.pbip` + `.Report` + `.SemanticModel`. `pbir validate --all` passes (cosmetic overlap / undersize warnings only — the KPI frame deliberately sits behind the cards).

First open in Power BI Desktop: Transform data → set `pDataFolder` to `data/curated` → Close & Apply · View ▶ Themes ▶ Browse → `powerbi/theme/theme.json` · Model view → mark `dim_date` as a date table (`date`).

Semantic model

19 imported curated tables + `dim_month` (month spine) + `Funnel Stage` + `_Measures` (62 measures). Relationships per `docs/05_data_model.md`: day-grain facts → `dim_date[date_key]`; month-grain facts → `dim_month[month_key]`; the member hub → `dim_member`; the three experiment facts → `dim_experiment_arm` / `dim_experiment_stratum`. Auto date/time off; implicit measures discouraged.

The four pages

#	Page	Question	KPI row (6 cards)	Content
1	Executive Summary	Program health, and did Flash Redemption work?	Active Members · Quarterly Redemption Rate (vs target) · Burn/Earn Ratio · Point Liability (adj) · Flash Redemption Lift · Engagement Lapse Rate	SkyPoints issued vs redeemed (line) · Exec Insight narrative card · effect-by-tier bar · liability trajectory (line)
2	Member Tiers & Value	Where is value concentrated, how big is the breakage liability?	Members · Active Rate · Elite Share · Revenue/Active Member · Card Penetration · Avg Points Balance	Active members by tier · revenue per member by tier · issued by earn source · tier upgrades vs downgrades · liability gross vs breakage-adjusted
3	A/B Test Results	Is the effect real, safe on guardrails, consistent?	Subjects · Control Rate · Treatment Rate · Lift (pp) · Significance · SRM (treatment share)	Redemption rate by arm × tier · treatment effect by tier (heterogeneity) · weekly redeemers by arm (novelty check) · guardrail table by arm
4	Redemption & Engagement Funnel	Where do members fall out, and who is about to lapse?	Ever-Redeemed Share · Quarterly Redemption Rate (vs target) · Redeeming Members · Account Lapse Rate · Engagement Lapse Rate · Reactivations	redemption funnel · redeemed by reward category · redemption rate vs plan (line) · engagement lapse rate by tier

Slicers on every page: **Analysis period** (`dim_month[month_name]`), **Experiment arm**, **Tier stratum**. KPI deltas are green > 0 / red ≤ 0 , inverted for the lower-is-better cards (breakage, lapse, liability, avg balance).

Known lite-scope gaps (not polished)

- Visual overlap / undersize warnings from `pbir validate` — the generated layout stacks the KPI frame behind the cards; z-order is not pinned per visual.
- No screenshot-verification loop (needs a live Desktop session).
- Theme is applied manually on first open (`pbir` cannot apply an arbitrary theme file from the CLI in this version).
- `Exec Insight` is a dynamic text measure (no Smart Narrative visual).

AeroVanti SkyPoints — Build Walkthrough

A running narrative of the build. Each entry: what was done, why, where to look. Newest at the bottom.

Phase 0 — Scope & scaffold (2026-09-02)

- Locked scope in `docs/00_scope.md`. Key calls: analysis-heavy (Phase 10 primary), Power BI "lite" (4 pages), reduced ETL (6 source systems, 6 DQ classes), BRL only, 120k enrolled members, 15k-member experiment.
- Created the folder tree under `PBI Pessoa1/AeroVanti-SkyPoints-Analytics/`.

Phase 1 — Company story & KPI framework (2026-09-02)

- `docs/01_business_context_kpis.md`. Fictional Brazilian domestic airline; SkyPoints relaunched Sep-2024 so the 24-month window is the whole modern program.
- Two point currencies (spendable SkyPoints, 24-month lot expiry; status Tier Points).
- Flash Redemption experiment fully specified: member-level, stratified-by-tier with Gold/Platinum oversampling, 50/50, primary = redemption rate, four guardrails, SRM + novelty + heterogeneity diagnostics, power/MDE rationale.

Phase 2 — Data architecture (2026-09-02)

- `docs/02_data_architecture.md`. Medallion raw -> staging -> curated. 6 simulated systems (SkyCore, BancoAV, Reserva, Aurora, Flesk, Loyalty Sub-Ledger). Curated star with pre-built experiment tables so Phase 10 goes straight to tests.

Phase 3 — ETL pipeline (2026-09-02)

3.1 The true effect (kept only in `etl/config.py::FLASH_EFFECT`)

Decided per instruction #3: real, non-trivial, not-obvious. Per-tier absolute lift in P(redeem in window): Blue +3.3pp, Silver +2.8pp, Gold +1.3pp, Platinum ~0. Design-weighted pooled ~+2.6pp; post-stratified to population mix ~+2.9pp (pooled understates because Blue is undersampled by the oversampling design). ~+1.4pp week-1 novelty bump decaying with $\tau=2.5$ weeks. Guardrails: revenue/member -0.4% (NI holds vs -3%), net liability cost/member favourable, average redemption *value* per redeeming member down ~R\$16 (shallower Flash redemptions — a watch item), 90-day post disengagement -1.6pp. A latent per-member "engagement" factor drives both redemption and retention so the *observational* redeemer/ non-redeemer lapse gap over-states causation; the experiment is the clean test.

3.2 Pipeline shape

`config.py` (one SEED, every tunable) · `utils.py` (rng, messy-export helpers, retrying writers) · `01_generate_raw.py` (builds the world in memory, emits source-shaped messy files) · `02_clean_stage.py` (typed `stg_*` + DQ fragments) · `03_build_curated.py` (star + experiment tables) · `dq_checks.py` (hard assertions + accounting identities + `dq_report.md`) · `run_pipeline.py` · `gen_data_dictionary.py`.

3.3 Fixes made during the build (this is where section-1-style guardrails come from)

- **Vectorise everything.** First cut had Python row loops in the point-lot engine, the raw `point_transactions` builder, and the member-month `months_since_last_activity` counter. At 120k members these ran for many minutes. Rewrote all three as vectorised numpy / groupby. 01 ~5 min, 03 ~5 min now.
- **Gzip files must be named `.csv.gz`.** `df.to_csv(path, compression="gzip")` writes gzip regardless of the name, but `pd.read_csv` only infers gzip from the `.gz` suffix -> staging blew up with `UnicodeDecodeError: 0x8b` (the gzip magic byte). Renamed the four large raw exports to `.csv.gz`.
- **Tier is a status property, not a derived flight count.** First design derived tier from trailing-12-month Tier Points. With a realistic mean of ~2-3 flights/year almost nobody reached Gold/Platinum, so the experiment's Gold/Platinum strata could not be filled (Platinum eligible pool = 5). Rebuilt: each member gets a `target_tier_rank` drawn to hit the 68/20/9/3 mix (tilted by engagement), and flight frequency is *boosted* by tier so revenue-by-tier still rises steeply.
- **Emit an initial "qualification" tier-change event.** The tier-change ledger only recorded month-over-month changes, so a member who started above Blue had no event and the curated fold-forward left them at Blue forever -> 10% snapshot-vs-

ledger "drift" that was actually a reconstruction bug. Added a Blue -> status event at enrolment.

- **Legacy opening balances need an issuance transaction.** Migrated members carry an opening SkyPoints balance (so the "members hoard points" starting condition and the mid-window breakage wave exist). It lived only in the lot matrix, so the transaction ledger didn't contain it and the points roll-forward identity broke once those lots expired. Added a one-off `legacy_migration` earn row at each migrated member's enrol month, and folded legacy into the sub-ledger's month-0 issuance.
- **pd.NA is not None.** The money-string parser did `x.strip()` inside a `.map`; pyarrow-backed string columns feed `pd.NA` (not Python `None`) to the mapper -> `AttributeError: 'NAType'`. Guarded with `pd.isna` and a list comprehension.
- **Windows console is cp1252.** `dq_checks.py` printed a Σ ; the script died on `UnicodeEncodeError` before writing the report. Added `sys.stdout.reconfigure(utf-8)` and kept console strings ASCII.
- **SKP_SCALE** (dev-only member-count scale) and **SKP_NO_CSV** (skip the curated CSV mirror) env knobs added to make calibration passes faster. Both default to the full build.

3.4 Calibration (7 full pipeline runs)

Tuned the redemption-hazard intercept A_0 ($-3.62 \rightarrow -3.25$) and the balance coefficient ($0.28 \rightarrow 0.18$) until the pre-experiment steady state sat mid-band. Two structural changes during calibration: - **Deterministic per-stratum conversion.** The first cut drew each treatment member's conversion from `Bernoulli(q)`, which added a second layer of Monte-Carlo noise on top of base-rate sampling — per-tier effects bounced ± 2 pp between runs. Switched to converting a **fixed count** `round(effect_pp * n_treatment)` of eligible non-redeemers. - **Blocked (pair-matched) randomisation.** Plain `permutation` produced a ~ 2.5 pp base-rate imbalance between arms in the Gold stratum on the project seed. Switched to ranking each stratum's chosen members by a pre-period redemption-propensity score, pairing adjacent ranks, and flipping a coin per pair. Arms are now balanced on prior behaviour by construction (Phase 10's balance check will show it) with real assignment randomness kept.

Final run (SEED 20260902) — all DQ hard + soft checks pass: - Program: 120k enrolled, ~ 109 k active @2026-08; tier mix 65/22/10/3; point liability R\$25.7M gross / R\$15.4M breakage-adjusted; burn/earn 30.4%; 228M pts (10.9% of issued) expired so far (legacy-lot wave — lifetime breakage trends to $\sim 40\%$). - Experiment: control redemption rate 12.8%, treatment 15.5%, **pooled lift +2.69 pp** ($n = 7,500/\text{arm}$). By stratum (lift pp): Blue +3.8, Silver +2.2, Gold +2.5, Platinum -0.5. Guardrails: revenue/member -0.1%, net liability cost/member -R\$0.81 (favourable), redemption value $\sim$ -R\$20 per *redeeming* member (shallower), 90-day disengagement -2.0 pp.

Phases 4-9 — docs + Power BI "lite" track (2026-09-02)

- **Phase 4:** `03_data_dictionary.md` auto-generated from `data/curated/` (20 tables); `04_etl_pipeline.md`, `05_data_model.md` (bus matrix + grains + 20 relationships), `08_data_quality.md` written.
- **Phase 5:** `assets/brand.py` (AeroVanti sky-blue / amber / steel palette), `assets/gen_logo.py` (mark + wordmark + 4 page-background PNGs), `powerbi/theme/theme.json` (adapted from the VoltEdge/Voxa theme, re-coloured), `docs/07_visual_identity.md`.
- **Phase 6:** `powerbi/gen_semantic_model.py` — generates the whole `.SemanticModel` (TMDL) from the curated Parquet schemas: 19 imported tables + a `dim_month` spine + Funnel Stage
- `_Measures`, Parquet import partitions via a `pDataFolder` parameter, 20 relationships. Adds two sentinel `dim_member` rows so non-member / orphan flight segments resolve.
- **Phase 7:** `etl/pbi_measures.py` — 62 measures in 10 display folders (Program, Earn & Burn, Breakage & Liability, Tiers/Commercial, Retention & Lapse, A/B Test, Time & Targets, Narrative, Funnel). The A/B group computes the lift, a two-proportion pooled z-stat, a Wald 95% CI on the lift, an SRM ratio, and the four guardrails. `Exec Insight` is a dynamic narrative text measure. `etl/gen_dax_doc.py` -> `docs/06_dax_measures.md`.
- **Phase 8:** `powerbi/gen_report_visuals.py` (4-page `_report_layout.json`), `_drive_pbir.py` (adapted from Voxa), `build_report.sh`. `pbir validate --all` **passes** (4 pages, 59 visuals, 128 fields; 99 cosmetic overlap/undersize warnings — the KPI frame sits behind the cards by design). Screenshot-verification skipped (needs a live Desktop). `docs/09_report_structure.md` written.
- **Phase 9:** this file.

Phase 10 — Data analysis (the primary deliverable) (2026-09-02)

`data_analysis/` — dual-track (pandas + DuckDB), 8 analyses, each self-checking parity inline; `parity/run_all.py` runs all and confirms every SQL/Python check passes.

- **01 anchor** — control 12.8% / treatment 15.5% redemption rate.
- **02 randomisation** — SRM $\chi^2=0$ ($p=1$) everywhere; max $|SMD|=0.039$. Blocked randomisation balances by construction.
- **03 primary** — +2.69 pp pooled, 95% CI [+1.58, +3.81], $z=4.74$, $p=2.2e-6$; post-stratified to the member base **+3.21 pp** [+2.20, +4.22]; covariate-adjusted AME +2.78 pp; power 0.97 at the +2.2 pp MDE.

- **04 guardrails** — revenue/member -0.09% (n.s.), **NI vs -3% NOT established** (one-sided $p=0.14$); net liability cost/member $-\text{R}\$0.81$ (n.s., favourable dir.); 90-day disengagement **-2.0 pp, $p=0.001$** ; value per redeeming member **$-\text{R}\$19$ (-6.8%), $p=0.001$** (shallower).
- **05 novelty** — weekly increment OLS slope -0.044 pp/wk, $p=0.017$; durable $\approx +1.5$ pp.
- **06 heterogeneity** — Blue $+3.8$ (sig) · Silver $+2.2$ (sig, marginal) · Gold $+2.5$ (n.s.) · Platinum -0.5 (n.s.); **arm×tier interaction** **$\text{LR } \chi^2=17.0$, $\text{df}=3$, $p=0.0007$** ; linear-in-rank interaction -0.172 , $p=0.0001$.
- **07 hoarding/liability** — 32% of active members ever redeemed (25% Blue); co-brand card = 55% of issuance; liability $\text{R}\$25.7\text{M}$ gross / $\text{R}\$15.4\text{M}$ adj; $\pm\text{R}\$2.6\text{M}$ per 10 pp of breakage assumption.
- **08 redeemer→retention** — engagement-lapse gap raw -10.5 pp, adjusted -6.0 pp ($p=1e-31$); latent-engagement caveat; the experiment's -2.0 pp 90-day guardrail is the clean causal signal.

Recommendation delivered: roll out Flash Redemption to **Blue + Silver**, hold Gold/Platinum, keep a revenue and a redemption-depth guardrail live; size the durable effect at $\approx +1.5-2$ pp. → this matches the baked `FLASH_EFFECT` (Blue/Silver real, Gold/Platinum ~ 0 , novelty decay, benign guardrails), detected blind.

Deliverables: `deliverables/board_summary.html` + `.pdf` (decision-first, AeroVanti identity);
`deliverables/deep_dive_00_index.md` + `deep_dive_A_ab_test_readout.md` + `deep_dive_B_program_health.md` (+ rendered `deep_dive.html` / `.pdf`). Built by `build_deliverables.py` (markdown → HTML → Chrome print-to-PDF).

Fix — Power BI load error: non-unique key on the "one" side (2026-09-02)

Opening the `.pbip` failed: *"a coluna 'month_key' na tabela 'dim_date' contém um valor duplicado '202701' e isso não é permitido ... para a chave primária de uma tabela"*. Cause: `gen_semantic_model.py` emitted an **inactive** `dim_month.month_key` → `dim_date.month_key` relationship "for optional drill". Power BI requires the "one" side of **any** relationship (active or not) to be unique, and `dim_date.month_key` has ~ 30 rows per month. Removed that relationship entirely — month-grain facts already join `dim_month` (unique `month_key`), day-grain facts join `dim_date[date_key]`; no `dim_month` ↔ `dim_date` link is needed. `docs/05` §4 updated. `pbir validate --all` still passes.

Fix — first-open in Desktop: theme import + blank/errored visuals (2026-09-02)

Data model loaded fine; three cosmetic/logic issues on first open: 1. **Theme import failed** ("problema com o arquivo JSON do tema"). The elaborate `visualStyles` block (adapted from the Voxa/VoltEdge theme) is rejected by this Desktop version. Stripped `theme.json` to an **import-safe minimal** theme (name, `dataColors`, core colours, semantic colours, `textClasses`) — the page-background PNGs carry the identity; visual styling falls back to Power BI defaults. 2. **KPI cards showed "(Em branco)"**. `dim_month` spanned to 2026-12, so `MAX(dim_month[month_key]) = 202612` — a month with no fact rows. Rebuilt `dim_month` to span **only the fact window, 2024-09 → 2026-08** (24 rows). 3. **Two visuals errored** ("Erro ao buscar os dados"). `Exp Redemption Rate` and `Exp Disengaged 90d % used` `AVERAGE( <boolean column> )`, which DAX rejects. Rewrote as `DIVIDE( CALCULATE( COUNTROWS(...), col = TRUE() ), COUNTROWS(...) )`. Every measure that chains off `Exp Redemption Rate` (control/treatment rate, lift, SRM, Exec Insight) is fixed by this.

Re-ran `gen_semantic_model.py` + `gen_dax_doc.py`; `pbir validate --all` passes.

Phases 11-12 — packaging (2026-09-02)

- **Phase 11:** `portfolio/gen_portfolio_images.py` → 5 gallery images (1600×820): cover, effect-by-tier, novelty decay, hoarding+liability, deliverables mock.
- **Phase 12:** `docs/build_docs_bundle.py` → `AeroVanti-SkyPoints-Documentation.pdf` (full, 2.2 MB) + `AeroVanti-SkyPoints-Summary.pdf` (0.7 MB, < 2 MB target). markdown → HTML → Chrome print-to-PDF.

3.5 Refinement noted for Phase 10 — "engagement lapse"

Realized 12-month **account** lapse (the doc-01 definition: no earn *and* no redemption) is low ($\sim 4\%$), because co-brand-card auto-earn keeps most accounts nominally active. The meaningful retention metric is **engagement lapse** — no *flight* and no *redemption* for 12 months — which Phase 10 computes from `fact_member_month` alongside the strict definition. `docs/01` §6 / §11.5 updated to introduce it.